

Controller Design for Flapping Flight

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1 Introduction

We propose a control strategy for flapping flight.

2 Model Description

2.1 Geometry and Aerodynamic Force

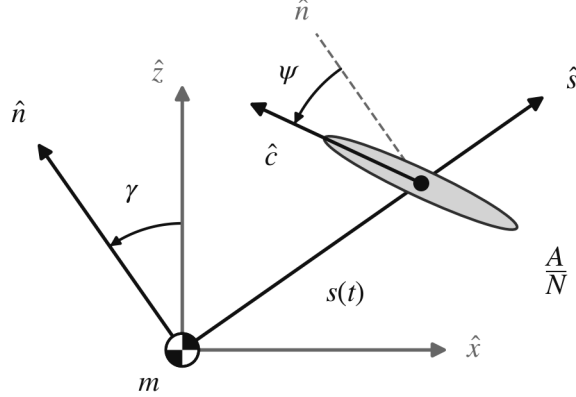


Figure 1: Model geometry. The stroke basis $(\hat{s}, \hat{n})$ is obtained by rotating the fixed frame basis $(\hat{x}, \hat{z})$ by the stroke plane angle γ . The lumped wing sits at position $s(t) \hat{s}$ relative to the point mass body, with chord direction $\hat{c}$ at pitch angle ψ from the stroke-plane normal $\hat{n}$. For simplicity, only one lumped wing is drawn.

The flapping flight dynamics model consists of a point mass subjected to its own weight $m\vec{g}$ and to the aerodynamic force $\vec{F}$ from N wing pairs. The point mass is free to move in the (x, z) plane, with the spatial direction $\hat{z}$ such that $\vec{g} = -g\hat{z}$, and $\hat{x}$ such that $(\hat{x}, \hat{z})$ forms a direct orthonormal basis. The body position is $\vec{x} = x\hat{x} + z\hat{z}$ and the body velocity is $\vec{u} = \dot{x}\hat{x} + \dot{z}\hat{z}$. The total wing area is A , and both wings of each pair are lumped into a single, massless surface with area A/N . We will refer to these surfaces as *lumped wings*, keeping in mind they each represent *both* wings of a left-right pair. The aerodynamic properties of a lumped wing are represented by the quasi-steady model of [1, 2]:

$$C_L(\alpha) = C_{L,0} \sin 2\alpha \quad (1)$$

$$C_D(\alpha) = C_{D,0} \cos^2 \alpha + C_{D,\frac{\pi}{2}} \sin^2 \alpha \quad (2)$$

where α is the angle of attack, and the parameter values are $C_{L,0} = 1.5$, $C_{D,0} = 0.1$, and $C_{D,\frac{\pi}{2}} = 2.0$. The model neglects any aerodynamic interaction between lumped wings. To express the lumped wing kinematics and aerodynamic force $\vec{F}$, it is convenient to introduce the stroke basis $(\hat{s}, \hat{n})$, obtained by rotating the fixed frame basis $(\hat{x}, \hat{z})$ by the *stroke plane angle* γ . That is

$$\hat{s} = \cos \gamma \hat{x} + \sin \gamma \hat{z} \quad (3)$$

$$\hat{n} = -\sin \gamma \hat{x} + \cos \gamma \hat{z} \quad (4)$$

Each lumped wing translates along the stroke axis $\hat{s}$, and $\hat{n}$ represents the stroke-plane normal. The lumped wing position relative to the point mass body is $\vec{s} = s(t)\hat{s}$, where $s(t)$ is the prescribed flapping coordinate. Because the wing is instantaneously at a distance $|s(t)|$ from the body, the aerodynamic force acts on the

body via a moment arm. However, in the present point mass model, there is no rotational degree of freedom, and the force is effectively considered to act through the center of mass.

Supposing γ stays constant during the wing stroke, the wing velocity relative to the body is $\vec{s} = \dot{s}(t)\hat{s}$. With the body velocity in the fixed frame (relative to still air) $\vec{u}$, the wing velocity relative to the air is $\vec{v} = \vec{s} + \vec{u}$. The aerodynamic force for wing pair i is given below (note the subscript i is omitted from quantities on the right-hand side for tidiness)

$$\vec{F}_i = \frac{1}{2}\rho \frac{A}{N} |\vec{v}|^2 [C_L(\alpha)\hat{\ell} + C_D(\alpha)\hat{d}] \quad (5)$$

where ρ is the air density and the drag direction vector $\hat{d}$ is defined as

$$\hat{d} = -\hat{v} = -\frac{\vec{v}}{|\vec{v}|} \quad (6)$$

and the lift direction vector $\hat{\ell}$ is defined as

$$\hat{\ell} = -\hat{d}^\perp \quad (7)$$

where $\cdot^\perp$ denotes counterclockwise rotation by $\pi/2$. As a result, $(\hat{\ell}, \hat{d})$ forms a direct orthonormal basis. To express the angle of attack α , it is convenient to define the chord direction vector $\hat{c}$, pointing from the trailing edge to the leading edge. Let $\psi(t)$ be the wing pitch angle, relative to the stroke-plane normal $\hat{n}$. The chord direction vector is

$$\hat{c} = -\sin\psi\hat{s} + \cos\psi\hat{n} \quad (8)$$

The geometry introduced thus far is illustrated in [Figure 1](#).

2.2 Angle of Attack and Aerodynamic Coefficients

The angle of attack α can be defined here as the signed angle from the velocity direction vector $\hat{v}$ to the chord direction vector $\hat{c}$. We show below that this definition is consistent with the definition of the lift direction vector $\hat{\ell}$, giving the correct sign for lift. With this definition,

$$\cos\alpha = \hat{v} \cdot \hat{c} \quad (9)$$

$$\sin\alpha = \hat{v} \times \hat{c} \quad (10)$$

where $\hat{v} = \vec{v}/|\vec{v}|$ and $\times$ denotes the 2D cross product, defined as $\vec{a} \times \vec{b} = a_1b_2 - a_2b_1$. The velocity composition, the angle of attack, and the resulting lift and drag are illustrated in [Figure 2](#). $C_L(\alpha)$ can be rewritten as a function of $\cos\alpha$ and $\sin\alpha$ only; indeed,

$$\begin{aligned} C_L(\alpha) &= C_{L,0} \sin 2\alpha \\ &= 2C_{L,0} \sin\alpha \cos\alpha \end{aligned} \quad (11)$$

C_L and C_D can then be expressed with simple vector math as follows

$$C_L = 2C_{L,0}(\hat{v} \cdot \hat{c})(\hat{v} \times \hat{c}) \quad (12)$$

$$C_D = C_{D,0}(\hat{v} \cdot \hat{c})^2 + C_{D,\frac{\pi}{2}}(\hat{v} \times \hat{c})^2 \quad (13)$$

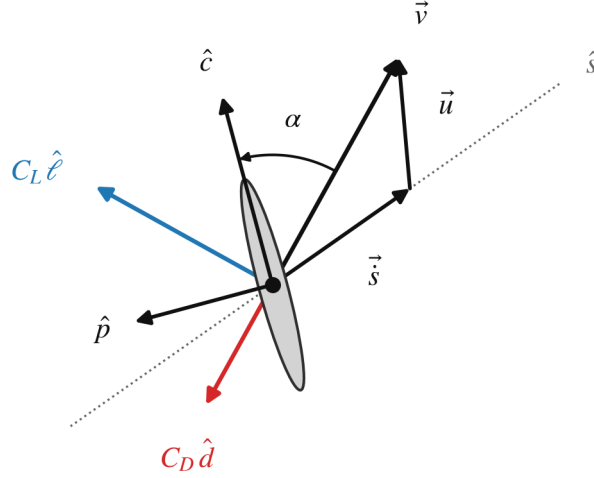


Figure 2: Lumped wing velocity triangle and aerodynamic force. The wing velocity relative to the air, $\vec{v} = \vec{s} + \vec{u}$, composes the flapping velocity $\vec{s} = s \hat{s}$ (along the stroke axis, dotted) with the body velocity $\vec{u}$. The angle of attack α is the signed angle from $\hat{v}$ to the chord direction $\hat{c}$, with $\hat{p} = \hat{c}^\perp$ the chord-normal direction, and the lift and drag contributions are drawn along $\hat{\ell}$ and $\hat{d}$ (as drawn, $0 < \alpha < \frac{\pi}{2}$, so $C_L > 0$).

Introducing the chord-normal direction $\hat{p} = \hat{c}^\perp$ and the wing velocity components along and normal to the chord, $v_c = \vec{v} \cdot \hat{c}$ and $v_p = \vec{v} \cdot \hat{p}$, we have $\hat{v} \cdot \hat{c} = v_c / |\vec{v}|$, $\hat{v} \times \hat{c} = -v_p / |\vec{v}|$, and $|\vec{v}|^2 = v_c^2 + v_p^2$, so the coefficients expand to the scalar form

$$C_L = -2C_{L,0} \frac{v_c v_p}{v_c^2 + v_p^2} \quad (14)$$

$$C_D = \frac{C_{D,0} v_c^2 + C_{D,\frac{\pi}{2}} v_p^2}{v_c^2 + v_p^2} \quad (15)$$

This treatment of C_L is the same as that of Andersen et al. [3], but with the opposite sign convention, α being defined from $\hat{c}$ to $\hat{v}$ in that work, whereas it is defined from $\hat{v}$ to $\hat{c}$ here. The lift and drag directions resulting from the above expressions are checked for correctness in all four quadrants of the angle of attack in Figure 3. Because $\hat{\ell}$ and $\hat{d}$ are constructed from $\hat{v}$ alone, the wing attitude enters only through C_L and C_D , so it suffices to check the coefficient signs quadrant by quadrant. Naturally, drag is positive along $\hat{d}$ for every α . The sign of C_L is that of $-v_c v_p$, which is that of $\sin 2\alpha$: positive in quadrants I and III, negative in II and IV, as drawn.

2.3 Prescribed Wing Kinematics

Wing motion is prescribed by the flapping coordinate $s(t)$ and the pitch angle $\psi(t)$. Flapping motion for each lumped wing is taken to be sinusoidal:

$$s(t) = s_0 \cos(\omega t + \phi) \quad (16)$$

where s_0 and ω are set parameters for all wing pairs, and ϕ is a phase parameter specific to a given wing pair, allowing for out-of-phase flapping of distinct wing pairs. For pitch, we consider two different kinematics. The first is sinusoidal:

$$\psi(t) = \psi_0 + \psi_1 \cos(\omega t + \phi - \delta_0) \quad (17)$$

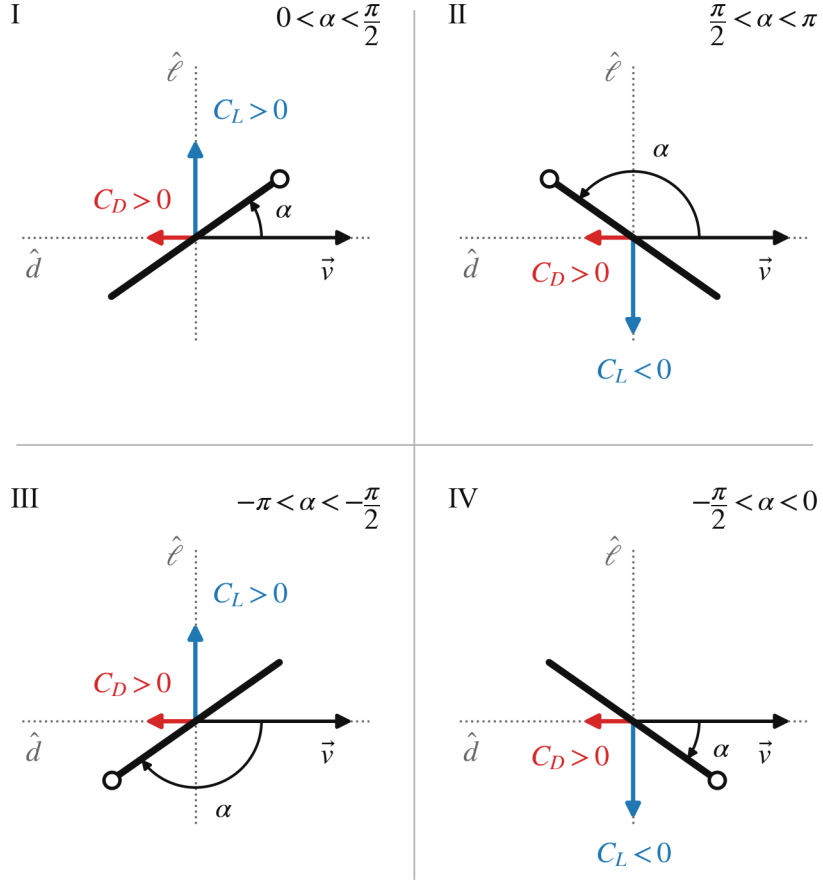


Figure 3: Lift and drag directions in the four quadrants of the angle of attack α , measured from the wing velocity direction $\hat{v}$ to the chord direction $\hat{c}$. Panels are labeled by quadrant (I–IV). Drag ($C_D > 0$) always opposes the wing velocity along $\hat{d}$, while the sign of C_L orients lift along $\pm \hat{\ell}$. The hollow circle marks the wing leading edge (the $\hat{c}$ end of the chord).

where ψ_0 (mean pitch), ψ_1 (pitch amplitude), and δ_0 (pitch phase lead) are set parameters for all wing pairs. The second kinematics law is simpler: through each half of the wing stroke, the wing pitch remains constant. The half-stroke along $-\hat{s}$ corresponds to $\omega t + \phi \in [0, \pi) \pmod{2\pi}$, and the half-stroke along $+\hat{s}$ to $\omega t + \phi \in [\pi, 2\pi) \pmod{2\pi}$. The simplified pitch kinematics are, then,

$$\psi(t) = \begin{cases} \psi_0 + \psi_1 & \text{if } \omega t + \phi \in [0, \pi) \pmod{2\pi} \\ \psi_0 - \psi_1 & \text{if } \omega t + \phi \in [\pi, 2\pi) \pmod{2\pi} \end{cases} \quad (18)$$

This can be considered an approximation of the sinusoidal law with $\delta_0 = \frac{\pi}{2}$. Both pitch kinematics laws are illustrated in Figure 4. As the figure makes apparent, $\delta_0 = \frac{\pi}{2}$ keeps the leading edge oriented into the wing motion throughout the cycle, with pitch extremes at mid-stroke and $\psi = \psi_0$ at stroke reversal. $\delta_0 = -\frac{\pi}{2}$ produces the opposite, adverse orientation, while $\delta_0 = 0$ and π place the pitch extremes at the reversals.

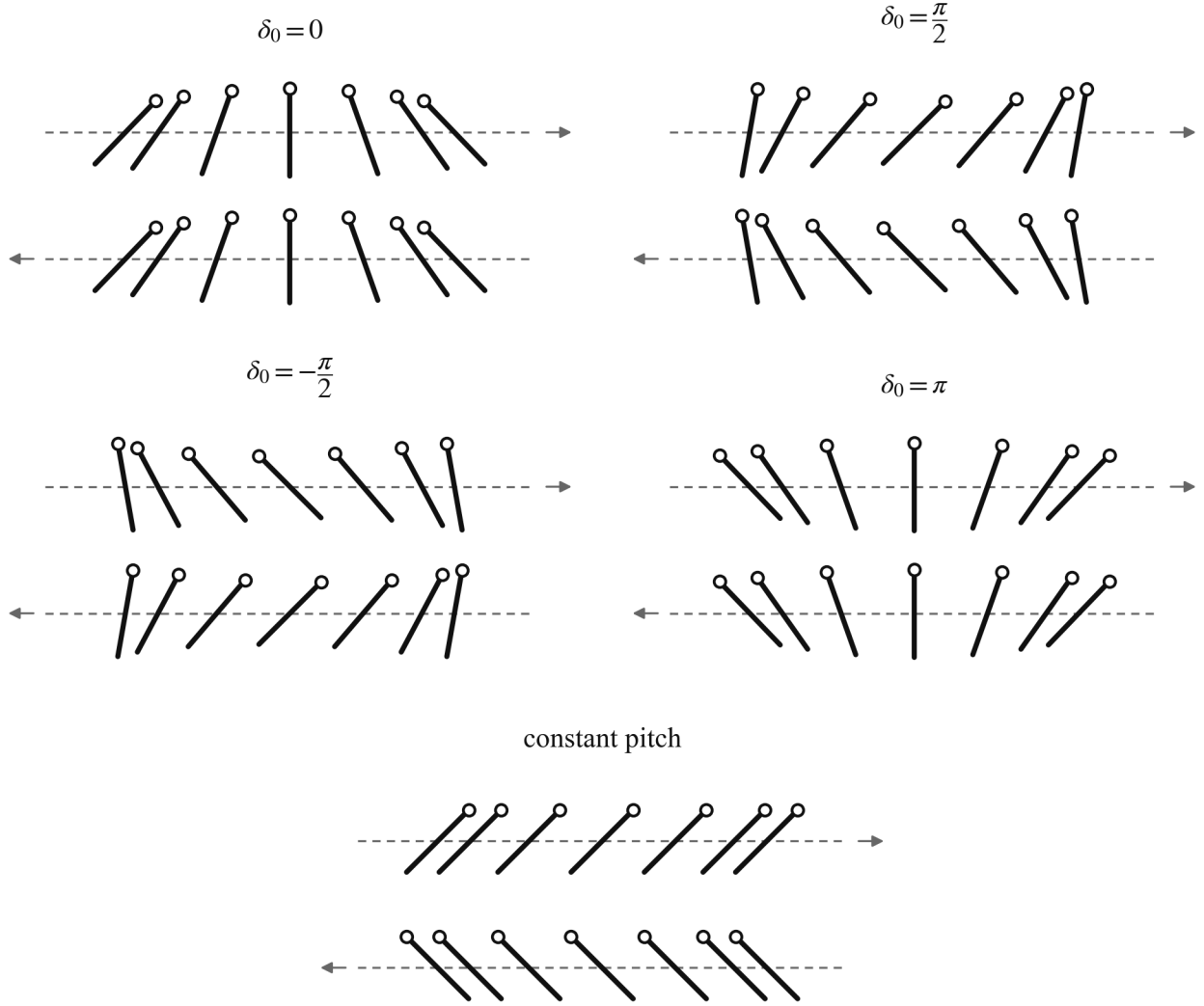


Figure 4: Wing motion through the two halves of the flapping cycle, for the sinusoidal pitch kinematics with $\delta_0 \in \{0, \frac{\pi}{2}, -\frac{\pi}{2}, \pi\}$ and for the constant-pitch simplification. All panels drawn with $\psi_0 = 0^\circ$ and $\psi_1 = 45^\circ$. The top row shows the half-stroke along $+\hat{s}$ ($\hat{s}$ pointing right) and the bottom row shows the half-stroke along $-\hat{s}$. Arrows indicate the direction of motion, the hollow circle marks the wing leading edge, and snapshots are taken at equal time increments, crowding near the stroke reversals where the flapping motion slows. The constant-pitch kinematics match the $\delta_0 = \pi/2$ kinematics well, except near stroke reversal.

2.4 Equation of Motion and Nondimensionalization

The point mass is subjected to its own weight and to the aerodynamic force from each wing pair, so the equation of motion is

$$m \frac{d^2 \vec{x}}{dt^2} = m \vec{g} + \frac{1}{2} \rho A \sum_{i=1}^N \frac{1}{N} |\vec{v}_i|^2 [C_{L,i} \hat{\ell}_i + C_{D,i} \hat{d}_i] \quad (19)$$

The equation can be nondimensionalized using the body length L as the length scale, and the gravitational time scale $\sqrt{L/g}$, giving an ‘acceleration scale’ g and a ‘velocity scale’ $\sqrt{gL}$. Introducing the $*$ superscript to denote nondimensionalized quantities, the equation of motion can be divided by mg and becomes

$$\begin{aligned} \ddot{x}^* \hat{x} + \ddot{z}^* \hat{z} &= -\hat{z} + \frac{1}{2} \frac{\rho AL}{m} \frac{1}{N} \sum_{i=1}^N \frac{1}{gL} |\vec{v}_i|^2 [C_{L,i} \hat{\ell}_i + C_{D,i} \hat{d}_i] \\ &= -\hat{z} + \frac{1}{2} \frac{\rho AL}{m} \frac{1}{N} \sum_{i=1}^N |\vec{v}_i^*|^2 [C_{L,i} \hat{\ell}_i + C_{D,i} \hat{d}_i] \end{aligned} \quad (20)$$

in which (dropping the i subscript for conciseness)

$$\vec{v}^* = \frac{1}{\sqrt{gL}} (\vec{u} + \vec{s}) \quad (21)$$

We introduce the dimensionless inverse wing loading A^*/m^* defined as

$$\frac{A^*}{m^*} = \frac{\rho AL}{m} \quad (22)$$

giving the final form of the dimensionless equation of motion

$$\ddot{x}^* \hat{x} + (\ddot{z}^* + 1) \hat{z} = \frac{1}{2} \frac{A^*}{m^*} \frac{1}{N} \sum_{i=1}^N |\vec{v}_i^*|^2 [C_{L,i} \hat{\ell}_i + C_{D,i} \hat{d}_i] \quad (23)$$

The wing kinematics themselves must also be nondimensionalized. Introducing $\tau^* = \omega t$ and $s_0^* = s_0/L$,

$$s^*(\tau^*) = s_0^* \cos(\tau^* + \phi) \quad (24)$$

$$\psi(\tau^*) = \psi_0 + \psi_1 \cos(\tau^* + \phi - \delta_0) \quad (25)$$

Finally, the dimensionless wing velocity along $\hat{s}$ relative to the body is

$$\begin{aligned} \dot{s}^*(\tau^*) &= \frac{1}{\sqrt{gL}} \frac{d}{dt} s_0 \cos(\omega t + \phi) \\ &= -\frac{1}{\sqrt{gL}} s_0 \omega \sin(\tau^* + \phi) \\ &= -s_0^* \omega^* \sin(\tau^* + \phi) \end{aligned} \quad (26)$$

with $\omega^* = \omega \sqrt{L/g}$. The dimensionless velocities can be used straightforwardly in place of the dimensional velocities in the expressions of [subsection 2.2](#) to obtain C_L and C_D . In the following, we deal exclusively with the dimensionless quantities.

3 Controller Design

3.1 Problem Statement

We wish to design a control strategy to achieve arbitrary trajectories in the two-dimensional plane, including prey pursuit trajectories. We assume that the flyer's own state is known using a variety of sensory inputs, so position and velocity are available as inputs to the controller. In addition, when prey is present in the environment, the bearing angle to the prey, as detected by the eyes, is also an available input. Because we are particularly interested in *dragonfly* flight, we make two design decisions such that the resulting flight characteristics resemble those of a dragonfly. First, hovering should be achieved with an inclined stroke plane (tilted from the horizontal), characterized by the parameter γ^{hover} . Second, when at speed, the stroke-plane normal should be approximately aligned with the direction of travel, as observed in experiments [4]. This last choice will simplify the analysis to follow, since any flow velocity component parallel to the stroke plane can be neglected. With these choices in mind, the contents of this section otherwise hold for flapping flight in general, as we do not assume specific morphological characteristics such as the number of wing pairs, size, etc.

3.2 Mean Aerodynamic Force

3.2.1 Definitions and Assumptions

To affect its trajectory, the flying body must modulate and direct the aerodynamic force acting on its wings. Because the aerodynamic force on a flapping wing is by nature highly unsteady, it is convenient to consider the wingbeat-cycle average of the aerodynamic force, summed over the wings, which we call the *mean aerodynamic force*, $\langle \vec{F}^* \rangle$. Throughout, $\langle \cdot \rangle$ denotes the average over one wingbeat cycle, $\langle f \rangle = \frac{1}{2\pi} \int_0^{2\pi} f d\tau^*$, evaluated at fixed body velocity. In reality, body velocity is not constant over a wingbeat cycle: there are (a) oscillations inherent to flapping, and there may be (b) a net change in body velocity between the start and end of the cycle. Assuming the oscillations are small compared to the characteristic wing velocity, and that any body velocity changes occur at a slow rate relative to the wingbeat period, the body velocity may be considered constant over a wingbeat cycle. The following assumes these conditions are met. Our objective is now to understand how $\langle \vec{F}^* \rangle$ depends on the wing kinematics at any given body velocity, so that we are able to formulate a control strategy.

As explained in [subsection 3.1](#), we are designing a controller under the assumption that the body velocity is normal to the stroke plane. Beyond morphology and kinematics, then, there is a single body velocity parameter $\langle u^* \rangle$, which is the cycle-averaged body velocity magnitude (the direction being set to the stroke-plane normal $\hat{n}$). It is convenient to introduce a similarity parameter in the form of the *advance ratio*, J , which is the ratio of the cycle-averaged body velocity to the characteristic wing velocity:

$$J = \frac{\langle u^* \rangle}{s_0^* \omega^*} \quad (27)$$

The instantaneous wing velocity $\vec{v}^*$ relative to the air has the stroke-plane normal and in-plane components

$$v_n^* = \langle u^* \rangle = s_0^* \omega^* J \quad (28)$$

$$v_s^* = -s_0^* \omega^* \sin \tau^* \quad (29)$$

with τ^* as introduced in [section 2](#); the $+\hat{s}$ half-stroke corresponds to $\sin \tau^* < 0$. Note that the phase term ϕ is dropped. Indeed, each wing pair carries the same area fraction $1/N$ and differs from the others only by its phase ϕ , which shifts its force history within the cycle without changing the cycle average at fixed body velocity. The per-pair averages are therefore identical, the sum over pairs and the $1/N$ factors cancel, and

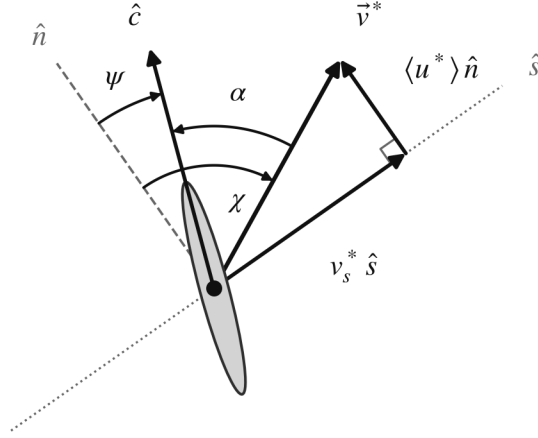


Figure 5: Stroke-frame velocity decomposition and flow angle. With the body velocity assumed normal to the stroke plane, the instantaneous wing velocity $\vec{v}^*$ has an in-plane flapping component $v_s^* \hat{s}$ and a normal component $\langle u^* \rangle \hat{n}$. The pitch angle ψ and the flow angle χ are measured from $\hat{n}$ with the same signed sense (positive when tilting toward $-\hat{s}$), and locate the chord direction $\hat{c}$ and the velocity direction $\hat{v}$, respectively, so the angle of attack is $\alpha = \psi - \chi$. As drawn, both ψ and χ are negative.

$\langle \vec{F}^* \rangle$ is computed below as for a single lumped wing of total area A , with ϕ dropped. Phase differences do shape the *instantaneous* force, which is smoother when the lumped wings beat out of phase, but not the mean force. The squared norm of the velocity is

$$(v^*)^2 = (s_0^* \omega^*)^2 [J^2 + \sin^2 \tau^*] \quad (30)$$

We introduce the signed angle χ from $\hat{n}$ to $\vec{v}^*$, defined as

$$\chi = -\tan^{-1} \frac{v_s^*}{v_n^*} = \tan^{-1} \frac{\sin \tau^*}{J} \quad (31)$$

Both ψ and χ are signed angles measured from $\hat{n}$ with the same sense (positive tilting counterclockwise, toward $-\hat{s}$). They locate the chord direction $\hat{c}$ and the velocity direction $\hat{v}$, respectively. The angle of attack as defined in [section 2](#) (the signed angle from $\hat{v}$ to $\hat{c}$) is therefore simply

$$\alpha = \psi - \chi = \psi - \tan^{-1} \frac{\sin \tau^*}{J} \quad (32)$$

The stroke-frame velocity decomposition and the angles ψ , χ , and α are illustrated in [Figure 5](#).

3.2.2 Hover Case ($J = 0$)

The mean aerodynamic force derivation is somewhat involved in the general case, so we first present the hovering case, which corresponds to $J = 0$. With $J = 0$, the flow angle saturates at $\chi = \frac{\pi}{2} \frac{\sin \tau^*}{|\sin \tau^*|}$ and the angle of attack simplifies to

$$\alpha = \psi - \frac{\pi}{2} \frac{\sin \tau^*}{|\sin \tau^*|} \quad (33)$$

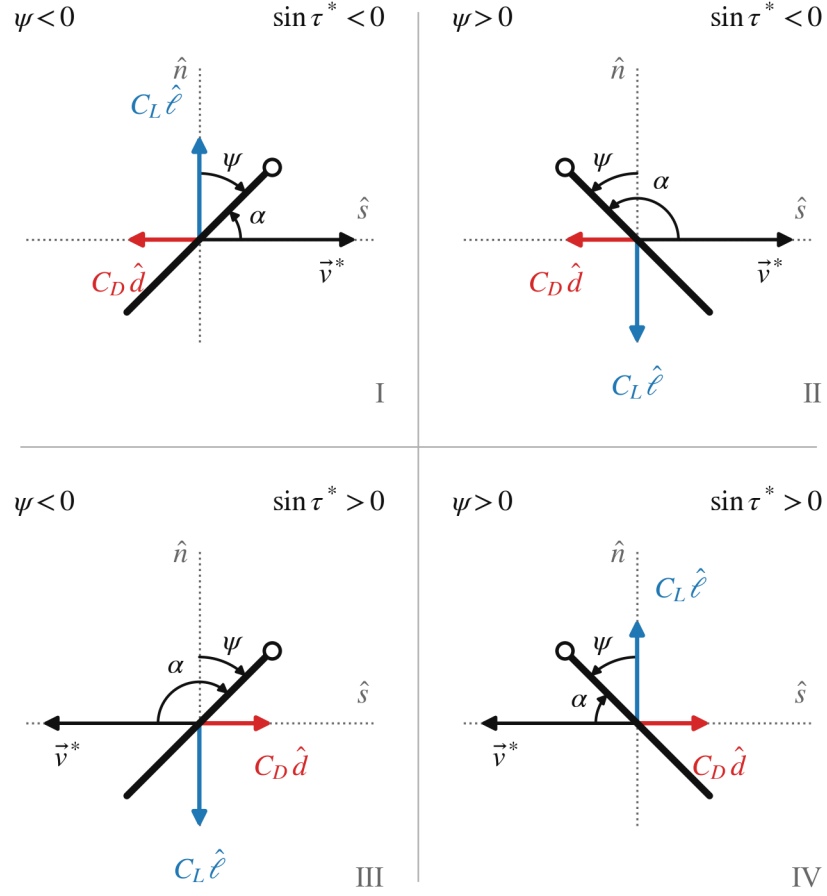


Figure 6: Lift and drag directions over the hover wingbeat ($J = 0$), for the four sign combinations of the pitch angle ψ and the stroke phase $\sin \tau^*$, drawn in the stroke frame. The wing velocity $\vec{v}^*$ is purely in-plane, so drag is purely in-plane, and lift is purely out-of-plane. The hollow circle marks the leading edge. The pitch angle ψ , measured from $\hat{n}$, and the angle of attack α of Equation 33, measured from $\hat{v}$, are marked as in Figure 5. α quadrants I–IV are marked as in Figure 3. The “useful” quadrants (lift along $\hat{n}$) are I and IV.

Using the appropriate trigonometric identities, the aerodynamic coefficients (defined in Equation 1 and Equation 2) are

$$C_L = -C_{L,0} \sin 2\psi \quad (34)$$

$$C_D = \frac{C_{D,\frac{\pi}{2}} + C_{D,0}}{2} + \frac{C_{D,\frac{\pi}{2}} - C_{D,0}}{2} \cos 2\psi \quad (35)$$

both independent of the stroke direction; the direction dependence resides entirely in the force direction vectors. Because the wing velocity relative to the air is purely in-plane (along $\hat{s}$), the in-plane aerodynamic force component is purely drag, and the out-of-plane component (along $\hat{n}$) is purely lift: the force directions are $\hat{d} = \frac{\sin \tau^*}{|\sin \tau^*|} \hat{s}$ and $\hat{\ell} = -\hat{d}^\perp = -\frac{\sin \tau^*}{|\sin \tau^*|} \hat{n}$, alternating with the stroke direction, as shown in Figure 6. The

mean aerodynamic force components are

$$\begin{aligned}\langle F_n^* \rangle &= -\frac{1}{2} \frac{A^*}{m^*} (s_0^* \omega^*)^2 \langle C_L \sin \tau^* | \sin \tau^* \rangle \\ &= \frac{1}{2} \frac{A^*}{m^*} (s_0^* \omega^*)^2 C_{L,0} \langle \sin 2\psi \sin \tau^* | \sin \tau^* \rangle\end{aligned}\quad (36)$$

$$\begin{aligned}\langle F_s^* \rangle &= \frac{1}{2} \frac{A^*}{m^*} (s_0^* \omega^*)^2 \langle C_D \sin \tau^* | \sin \tau^* \rangle \\ &= \frac{1}{2} \frac{A^*}{m^*} (s_0^* \omega^*)^2 \frac{C_{D,\frac{\pi}{2}} - C_{D,0}}{2} \langle \cos 2\psi \sin \tau^* | \sin \tau^* \rangle\end{aligned}\quad (37)$$

Note that the constant drag term vanishes because $\langle \sin \tau^* | \sin \tau^* \rangle = 0$. We introduce the shorthand $\Delta C_D = C_{D,\frac{\pi}{2}} - C_{D,0}$. The expressions feature a common pre-factor we call q^* , defined as

$$q^* = \frac{1}{4} \frac{A^*}{m^*} (s_0^* \omega^*)^2 \quad (38)$$

We further introduce the normal and in-plane force coefficients $C_{F_n^*}$ and $C_{F_s^*}$, defined as

$$C_{F_n^*} = \frac{\langle F_n^* \rangle}{q^*} = 2C_{L,0} \langle \sin 2\psi \sin \tau^* | \sin \tau^* \rangle \quad (39)$$

$$C_{F_s^*} = \frac{\langle F_s^* \rangle}{q^*} = \Delta C_D \langle \cos 2\psi \sin \tau^* | \sin \tau^* \rangle \quad (40)$$

which we synthesize into the overall force magnitude coefficient

$$C_{F^*} = \sqrt{C_{F_n^*}^2 + C_{F_s^*}^2} \quad (41)$$

This coefficient is useful because it is only a function of the pitch kinematic parameters and the lift and drag coefficient parameters, and is independent of morphology and wingbeat frequency and amplitude, which appear only in the pre-factor q^* . By maximizing the value of this coefficient, given a constraint on the force direction required to achieve a given maneuver, the characteristic wingbeat velocity $s_0^* \omega^*$ can be minimized. With our parametrization of the pitch kinematics, and with the lift and drag coefficient parameters set, C_{F^*} and its (n, s) decomposition are functions of three variables: ψ_0 , ψ_1 , and δ_0 . We visualize C_{F^*} , $C_{F_n^*}$, and $C_{F_s^*}$ across slices of this 3D space in [Figure 7](#) and [Figure 8](#).

As panel (a) of [Figure 8](#) makes evident, significantly more force is produced when $\delta_0 = \pm 90^\circ$. Because $\delta_0 = -90^\circ$ results in the mean force pointing along $-\hat{n}$ when the mean pitch is neutral ($\psi_0 = 0^\circ$), we select $\delta_0 = +90^\circ$ as the operating point. This choice maximizes the mean force for a given (ψ_0, ψ_1) , and reduces the pitch parameter space to 2D. How then do we select ψ_0 and ψ_1 ? It is clear from panel (b) that a certain value of ψ_1 maximizes C_{F^*} for any given ψ_0 . We will call it ψ_1^{opt} . In addition, $\psi_0 = 0^\circ \pmod{90^\circ}$ maximizes C_{F^*} . Looking at panel (b) of [Figure 7](#), it is clear that the in-plane force component is zero when $\psi_0 = 0^\circ \pmod{90^\circ}$, so the force is purely normal to the stroke plane. In addition, from panel (a) of the same figure, we see that $\psi_0 = 0^\circ$ results in the mean force pointing towards $+\hat{n}$, while $\psi_0 = \pm 90^\circ$ results in the mean force pointing towards $-\hat{n}$. We select neutral pitch ($\psi_0 = 0^\circ$) as the reference, for consistency with the choice of $\delta_0 = 90^\circ$. In summary, the reference pitch kinematics

$$\psi_0 = 0^\circ \quad \psi_1 = \psi_1^{\text{opt}} \quad \delta_0 = 90^\circ \quad (42)$$

result in $C_{F^*} = C_{F^*}^{\text{max}}$, and $\langle \vec{F}^* \rangle$ pointing towards $+\hat{n}$.

These reference kinematics allow for *efficient* hover (in the sense that the characteristic wingbeat velocity $s_0^* \omega^*$ is minimized, because C_{F^*} is maximized) when the stroke plane is horizontal ($\gamma^{\text{hover}} = 0^\circ$). Indeed,

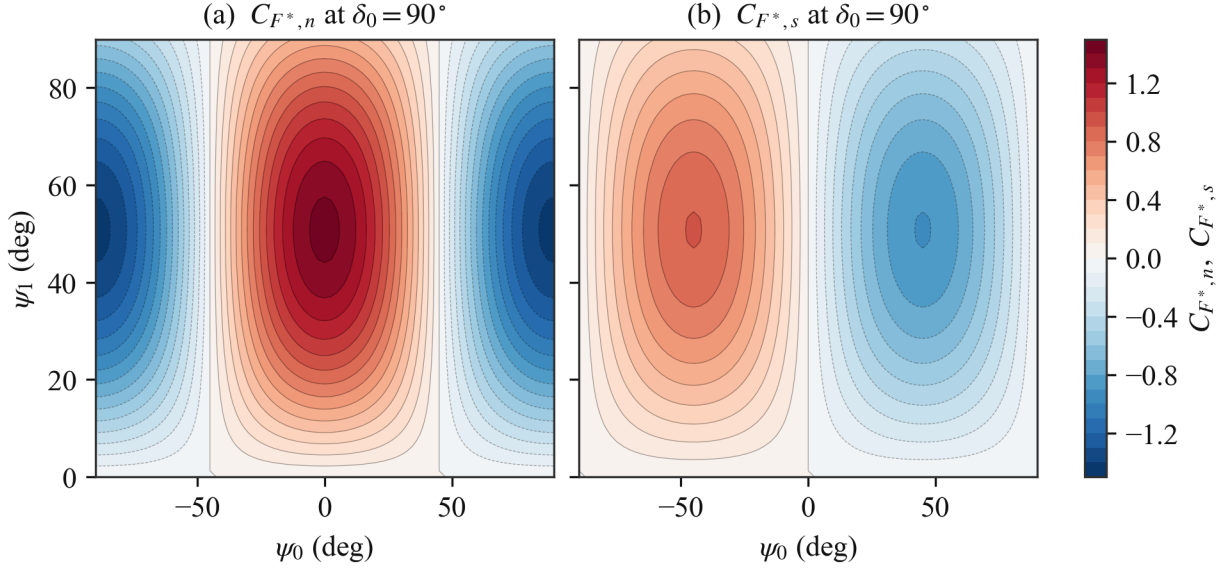


Figure 7: (a) Stroke-plane-normal (lift) component $C_{F_n}^*$ and (b) in-plane (drag) component $C_{F_s}^*$ of the force coefficient over the pitch plane at $\delta_0 = 90^\circ$. The force direction quadrant in the $(\hat{s}, \hat{n})$ plane can be inferred from the signs.

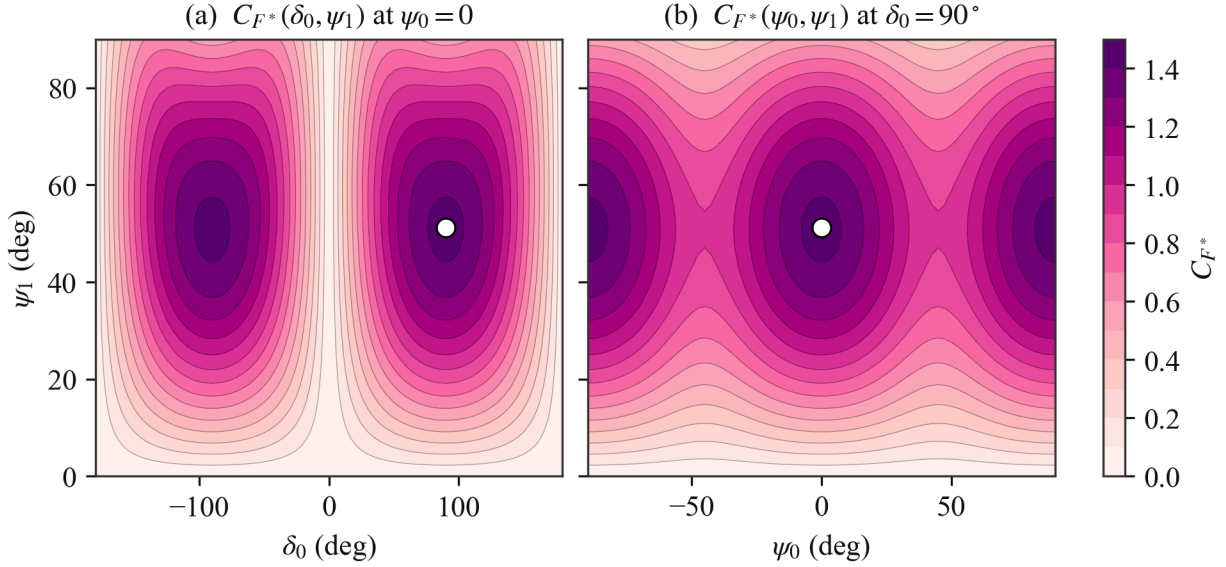


Figure 8: Dependence of C_{F^*} on pitch parameters at zero body velocity over (a) the (δ_0, ψ_1) plane at $\psi_0 = 0$ and (b) the (ψ_0, ψ_1) plane at $\delta_0 = 90^\circ$. Maximum shown as a white dot on both panels.

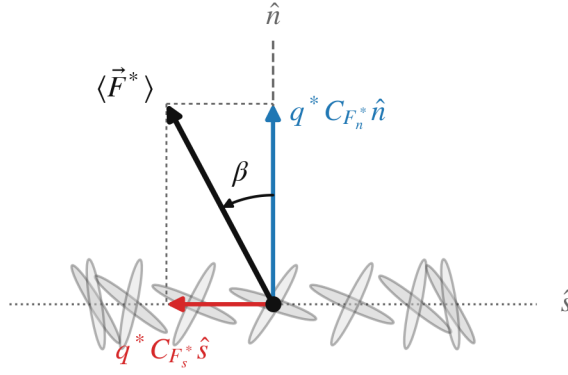


Figure 9: Definition of the mean-force direction angle β . The mean force $\langle \vec{F}^* \rangle = q^* (C_{F_n^*} \hat{n} + C_{F_s^*} \hat{s})$ is decomposed into its stroke-frame components (Equation 39, Equation 40). Here the positive mean pitch $\psi_0 > 0$ gives $C_{F_s^*} < 0$, so the force tilts counterclockwise off the normal and β is positive. The wing is drawn over one full wingbeat.

since the mean force vector points towards $+\hat{n}$, $+\hat{n}$ must be vertical and opposed to the direction of gravity. To model dragonfly hover specifically, at an inclined stroke plane ($\gamma^{\text{hover}} \neq 0^\circ$), the mean force vector must have an in-plane component. Looking at panel (b) of Figure 7, it is clear that this can be achieved by changing ψ_0 , away from the symmetric 0° point. For convenience, we introduce β , the signed angle from the stroke-plane normal $\hat{n}$ to the mean force vector. Since the mean force is $\langle \vec{F}^* \rangle = q^* (C_{F_n^*} \hat{n} + C_{F_s^*} \hat{s})$, β can be expressed from the $C_{F_n^*}$ and $C_{F_s^*}$ coefficients (Equation 39 and Equation 40) as

$$\beta = -\tan^{-1} \frac{C_{F_s^*}}{C_{F_n^*}} = -\tan^{-1} \left[\frac{\Delta C_D}{2C_{L,0}} \frac{\langle \cos 2\psi \sin \tau^* | \sin \tau^* \rangle}{\langle \sin 2\psi \sin \tau^* | \sin \tau^* \rangle} \right] \quad (43)$$

β carries the same signed sense as ψ and χ (positive counterclockwise, from $\hat{n}$ toward $-\hat{s}$), hence the minus sign (see Figure 9). We can simplify the $\langle \cos 2\psi \sin \tau^* | \sin \tau^* \rangle / \langle \sin 2\psi \sin \tau^* | \sin \tau^* \rangle$ ratio in the above expression. With $\delta_0 = 90^\circ$, the pitch kinematics reduce to $\psi = \psi_0 + \psi_1 \sin \tau^*$. Using the appropriate trigonometric identities,

$$\cos 2\psi = \cos 2\psi_0 \cos (2\psi_1 \sin \tau^*) - \sin 2\psi_0 \sin (2\psi_1 \sin \tau^*) \quad (44)$$

$$\sin 2\psi = \sin 2\psi_0 \cos (2\psi_1 \sin \tau^*) + \cos 2\psi_0 \sin (2\psi_1 \sin \tau^*) \quad (45)$$

Under the half-period shift $\tau^* \rightarrow \tau^* + \pi$, the factors $\sin \tau^* | \sin \tau^* \rangle$ and $\sin (2\psi_1 \sin \tau^*)$ change sign while $\cos (2\psi_1 \sin \tau^*)$ is unchanged, so the $\cos (2\psi_1 \sin \tau^*)$ terms average to zero, leading to

$$\langle \cos 2\psi \sin \tau^* | \sin \tau^* \rangle = -\sin 2\psi_0 \langle \sin (2\psi_1 \sin \tau^*) \sin \tau^* | \sin \tau^* \rangle \quad (46)$$

$$\langle \sin 2\psi \sin \tau^* | \sin \tau^* \rangle = \cos 2\psi_0 \langle \sin (2\psi_1 \sin \tau^*) \sin \tau^* | \sin \tau^* \rangle \quad (47)$$

The common $\langle \sin (2\psi_1 \sin \tau^*) \sin \tau^* | \sin \tau^* \rangle$ factor cancels out in the ratio, giving simply

$$\frac{\langle \cos 2\psi \sin \tau^* | \sin \tau^* \rangle}{\langle \sin 2\psi \sin \tau^* | \sin \tau^* \rangle} = -\tan 2\psi_0 \quad (48)$$

and Equation 43 becomes

$$\beta = \tan^{-1} \left[\frac{\Delta C_D}{2C_{L,0}} \tan 2\psi_0 \right] \quad (49)$$

Importantly, β is independent of ψ_1 , and is only controlled by ψ_0 . It follows that ψ_0 is an appropriate control variable for the mean force direction relative to the stroke-plane normal $\hat{n}$. β is shown as a function of ψ_0 in Figure 10. As discussed above, C_{F^*} becomes less than its maximum when $\psi_0 \neq 0^\circ \pmod{90^\circ}$, away from the reference pitch kinematics, so it is shown alongside β . In the $\Delta C_D / (2C_{L,0}) \rightarrow 1$ limit, the relation between β

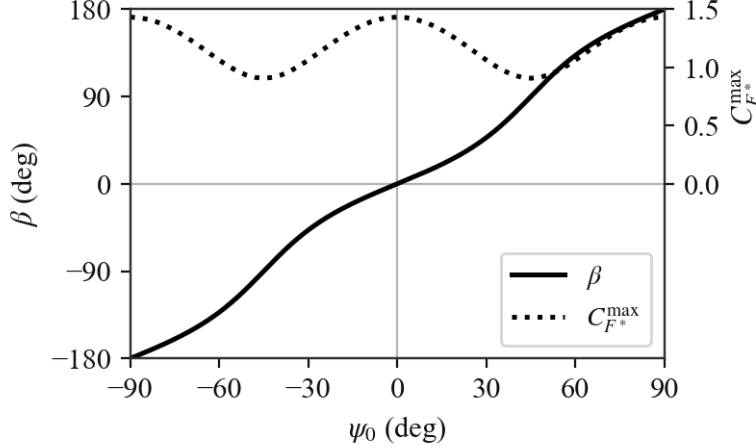


Figure 10: Effect of ψ_0 on force direction angle β and force coefficient C_{F^*} .

and ψ_0 can be approximated as $\beta \approx 2\psi_0$. This holds only roughly here because $\Delta C_D / (2C_{L,0}) = 1.9/3 \approx 0.63$, but the trend is the same, as can be seen in the figure. To achieve hover at an inclined stroke plane with tilt angle γ^{hover} , the mean force vector angle must be $\beta = -\gamma^{\text{hover}}$. This requires $\psi_0 = \psi_0^{\text{hover}}$, given by Equation 49 (specifically the inverse relation) when $\beta = -\gamma^{\text{hover}}$. The two hover configurations are illustrated in Figure 11.

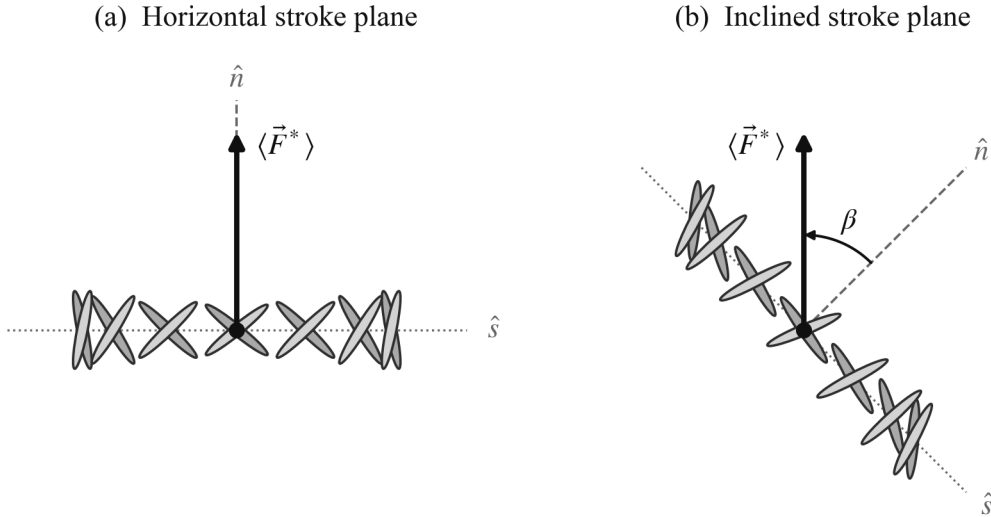


Figure 11: Hover at (a) a horizontal stroke plane and (b) an inclined stroke plane (drawn here with $\gamma < 0$). The wing is drawn over one full wingbeat at equal time increments, with the $-\hat{s}$ half-stroke behind, in darker gray, and the $+\hat{s}$ half-stroke in front. In (a), the reference pitch kinematics ($\psi_0 = 0^\circ$) set the mean force $\langle \vec{F}^* \rangle$ vertically along $\hat{n}$. In (b), the mean force remains vertical while $\hat{n}$ is tilted because ψ_0^{hover} is set to rotate $\langle \vec{F}^* \rangle$ by $\beta = -\gamma^{\text{hover}}$ using the inverse of Equation 49.

Now that we have narrowed down the pitch kinematics for hover, we step back to consider the flapping kinematics as well, governed by parameters s_0^* (amplitude) and ω^* (frequency). The condition for hover on the mean force magnitude is

$$C_{F^*} q^* = 1 \implies (s_0^* \omega^*)^2 = \frac{4}{C_{F^*} \frac{A^*}{m^*}} \quad (50)$$

We prefer to consider ω^* as a fixed parameter rather than a control variable, leaving s_0^* as the primary control for the force amplitude. The hover value is given by

$$(s_0^*)^{\text{hover}} = \frac{1}{\omega^*} \sqrt{\frac{4}{C_{F^*}^{\text{hover}} \frac{A^*}{m^*}}} \quad (51)$$

The hover kinematic parameters are summarized in Table 1. The control variables are s_0^* and ψ_0 .

Table 1: Wing kinematic parameters for hover

Parameter	Value	Depends on
s_0^*	$(s_0^*)^{\text{hover}}$	$\frac{A^*}{m^*}$ ω^* γ^{hover} $C_{L,0}$ ΔC_D
ψ_0	ψ_0^{hover}	γ^{hover} $C_{L,0}$ ΔC_D
ψ_1	ψ_1^{opt}	None
δ_0	90°	None

3.2.3 General Case ($J \geq 0$)

For $J \neq 0$, the wing velocity relative to the air is no longer purely in-plane, so lift and drag each contribute to both force components. In terms of the angle χ , the wing velocity direction is $\hat{v} = \cos \chi \hat{n} - \sin \chi \hat{s}$. Drag acts along $-\hat{v}$, and lift acts along the perpendicular direction

$$\hat{\ell} = -\hat{d}^\perp = -\cos \chi \hat{s} - \sin \chi \hat{n} \quad (52)$$

recovering the hover directions at $\chi = \pm \frac{\pi}{2}$. The instantaneous force is

$$\vec{F}^* = \frac{1}{2} \frac{A^*}{m^*} (v^*)^2 (C_L \hat{\ell} - C_D \hat{v}) \quad (53)$$

with components

$$F_n^* = -\frac{1}{2} \frac{A^*}{m^*} (v^*)^2 (\sin \chi C_L + \cos \chi C_D) \quad (54)$$

$$F_s^* = -\frac{1}{2} \frac{A^*}{m^*} (v^*)^2 (\cos \chi C_L - \sin \chi C_D) \quad (55)$$

Substituting $\cos \chi = J / \sqrt{J^2 + \sin^2 \tau^*}$, $\sin \chi = \sin \tau^* / \sqrt{J^2 + \sin^2 \tau^*}$, and the expression for $(v^*)^2$ (Equation 30), we get

$$F_n^* = -\frac{1}{2} \frac{A^*}{m^*} (s_0^* \omega^*)^2 \sqrt{J^2 + \sin^2 \tau^*} (\sin \tau^* C_L + J C_D) \quad (56)$$

$$F_s^* = -\frac{1}{2} \frac{A^*}{m^*} (s_0^* \omega^*)^2 \sqrt{J^2 + \sin^2 \tau^*} (J C_L - \sin \tau^* C_D) \quad (57)$$

From $\alpha = \psi - \chi$ (Equation 32), the aerodynamic coefficients (Equation 1 and Equation 2) expand as

$$C_L = C_{L,0} (\cos 2\chi \sin 2\psi - \sin 2\chi \cos 2\psi) \quad (58)$$

$$C_D = \bar{C}_D - \frac{\Delta C_D}{2} (\cos 2\chi \cos 2\psi + \sin 2\chi \sin 2\psi) \quad (59)$$

where we define $\bar{C}_D \equiv (C_{D, \frac{\pi}{2}} + C_{D,0})/2$ for brevity, and where the double angles are smooth functions of the wing phase:

$$\cos 2\chi = \frac{J^2 - \sin^2 \tau^*}{J^2 + \sin^2 \tau^*} \quad \sin 2\chi = \frac{2J \sin \tau^*}{J^2 + \sin^2 \tau^*} \quad (60)$$

Substituting into the force components and averaging, the force coefficients are

$$C_{F_n^*} = -2 \left\langle \frac{[C_{L,0} (J^2 - \sin^2 \tau^*) - \Delta C_D J^2] \sin \tau^* \sin 2\psi}{\sqrt{J^2 + \sin^2 \tau^*}} \right\rangle + 2 \left\langle \frac{J [2C_{L,0} \sin^2 \tau^* + \frac{\Delta C_D}{2} (J^2 - \sin^2 \tau^*)] \cos 2\psi}{\sqrt{J^2 + \sin^2 \tau^*}} \right\rangle - 2J\bar{C}_D \langle \sqrt{J^2 + \sin^2 \tau^*} \rangle \quad (61)$$

$$C_{F_s^*} = -2 \left\langle \frac{J [C_{L,0} (J^2 - \sin^2 \tau^*) + \Delta C_D \sin^2 \tau^*] \sin 2\psi}{\sqrt{J^2 + \sin^2 \tau^*}} \right\rangle + 2 \left\langle \frac{[2C_{L,0} J^2 - \frac{\Delta C_D}{2} (J^2 - \sin^2 \tau^*)] \sin \tau^* \cos 2\psi}{\sqrt{J^2 + \sin^2 \tau^*}} \right\rangle \quad (62)$$

Note that the constant ($\bar{C}_D$) drag term of $C_{F_s^*}$ vanishes because $\langle \sin \tau^* \sqrt{J^2 + \sin^2 \tau^*} \rangle = 0$. Setting $J = 0$ recovers Equation 39 and Equation 40, as expected. At the operating point $\delta_0 = 90^\circ$, $\psi = \psi_0 + \psi_1 \sin \tau^*$, and the parity argument of the hover case applies unchanged: under $\tau^* \rightarrow \tau^* + \pi$, the bracketed polynomials and $\sqrt{J^2 + \sin^2 \tau^*}$ are invariant, so expanding $\sin 2\psi$ and $\cos 2\psi$ and discarding the vanishing averages leaves

$$C_{F_n^*} = 2 I_n(J, \psi_1) \cos 2\psi_0 - 2J\bar{C}_D I_0(J) \quad (63)$$

$$C_{F_s^*} = -2 I_s(J, \psi_1) \sin 2\psi_0 \quad (64)$$

with

$$I_0 = \langle \sqrt{J^2 + \sin^2 \tau^*} \rangle \quad (65)$$

$$I_n = \left\langle \frac{J [2C_{L,0} \sin^2 \tau^* + \frac{\Delta C_D}{2} (J^2 - \sin^2 \tau^*)] \cos (2\psi_1 \sin \tau^*)}{\sqrt{J^2 + \sin^2 \tau^*}} \right\rangle - \left\langle \frac{[C_{L,0} (J^2 - \sin^2 \tau^*) - \Delta C_D J^2] \sin \tau^* \sin (2\psi_1 \sin \tau^*)}{\sqrt{J^2 + \sin^2 \tau^*}} \right\rangle \quad (66)$$

$$I_s = \left\langle \frac{J [C_{L,0} (J^2 - \sin^2 \tau^*) + \Delta C_D \sin^2 \tau^*] \cos (2\psi_1 \sin \tau^*)}{\sqrt{J^2 + \sin^2 \tau^*}} \right\rangle + \left\langle \frac{[2C_{L,0} J^2 - \frac{\Delta C_D}{2} (J^2 - \sin^2 \tau^*)] \sin \tau^* \sin (2\psi_1 \sin \tau^*)}{\sqrt{J^2 + \sin^2 \tau^*}} \right\rangle \quad (67)$$

The angle from the stroke-plane normal to the mean force (Equation 43) becomes

$$\beta = \tan^{-1} \left[\frac{I_s \sin 2\psi_0}{I_n \cos 2\psi_0 - J \bar{C}_D I_0} \right] \quad (68)$$

In the limit $J \rightarrow 0$, the I_0 term vanishes and

$$I_n \rightarrow C_{L,0} \langle \sin(2\psi_1 \sin \tau^*) \sin \tau^* | \sin \tau^* | \rangle \quad I_s \rightarrow \frac{\Delta C_D}{2} \langle \sin(2\psi_1 \sin \tau^*) \sin \tau^* | \sin \tau^* | \rangle \quad (69)$$

recovering Equation 49. The overall force coefficient C_{F^*} is shown in Figure 12 for values of J from 0 to 1, at $\delta_0 = 90^\circ$. Highlighted on each panel is the reference point from the hover plot of Figure 8, tracked as it turns from a global maximum at $J = 0$ to a local maximum, and eventually to a saddle point. This point represents the maximum force available in the $+\hat{n}$ direction for a given J . The region of space within the dotted contour around the reference point corresponds to $C_{F_n^*} > 0$. Within this region, the stroke-plane-normal component of the mean force points towards $+\hat{n}$.

The values of C_{F^*} and ψ_1^{opt} at the reference point are shown as functions of J in Figure 13. As the figure makes clear, $\psi_1^{\text{opt}}(J)$ decreases as J increases. Recall that $\alpha = \psi - \chi$ and consider the half-stroke along $+\hat{s}$, as pictured in Figure 5. Mid-stroke, when $\sin \tau^* = \mp 1$, $\chi = \mp \tan^{-1}(1/J)$ ($-$ when the stroke is along $+\hat{s}$, $+$ when it is along $-\hat{s}$). This corresponds to a rotation of $\vec{v}^*$ towards the stroke-plane normal $\hat{n}$ by an angle of magnitude $\pi/2 - \tan^{-1}(1/J) = \tan^{-1} J$. This suggests the following trend for $\psi_1^{\text{opt}}(J)$:

$$\psi_1^{\text{opt}}(J) \approx \psi_1^{\text{opt}}(0) - \tan^{-1} J \quad (70)$$

which is also shown in Figure 13, and does appear to be the leading-order effect.

Finally, Figure 14 shows β as a function of ψ_0 for J from 0 to 1. Specifically, since β now depends on ψ_1 in addition to ψ_0 (see Equation 68), each line is obtained using the ψ_1 value adapted to the value of J (as shown in Figure 13). As J is increased, there is a clear steepening of the slope around $\psi_0 = 0$, meaning control authority is significantly increased.

We summarize the kinematic parameters for the general case in Table 2. Since there is no requirement for steady flight as in hover, no specific values for the control variables s_0^* and ψ_0 are prescribed; however, the relevant variables that would inform the choice of the values are still listed.

Table 2: Wing kinematic parameters (general case)

Parameter	Value	Depends on						
s_0^*	—	$\frac{A^*}{m^*}$	ω^*	γ	$C_{L,0}$	ΔC_D	$\bar{C}_D$	J
ψ_0	—			γ	$C_{L,0}$	ΔC_D	$\bar{C}_D$	J
ψ_1	$\psi_1^{\text{opt}}(J)$				J	$C_{L,0}$	ΔC_D	
δ_0	90°							None

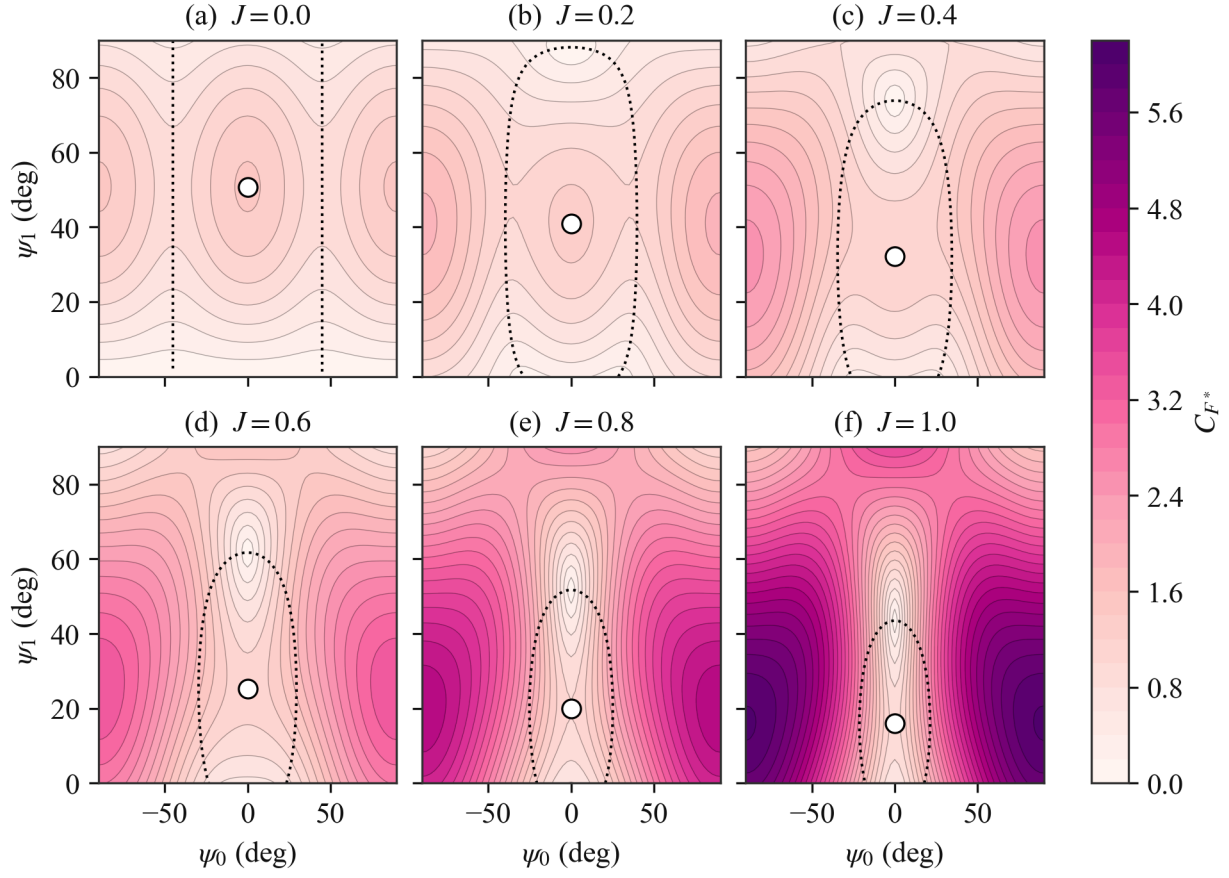


Figure 12: Force coefficient C_{F^*} over the pitch plane (ψ_0, ψ_1) at $\delta_0 = 90^\circ$, for increasing advance ratio J . White dots track the hover optimum as it moves with J . It remains on the $\psi_0 = 0^\circ$ axis by symmetry while ψ_1 decreases, and it turns from a local maximum into a saddle point. The global maximum sits at large $|\psi_0|$, where the force is dominated by drag from the normal inflow and points along $-\hat{n}$. The dotted contour marks $C_{F_n^*} = 0$, the boundary between propulsive ($+\hat{n}$) and braking ($-\hat{n}$) mean force.

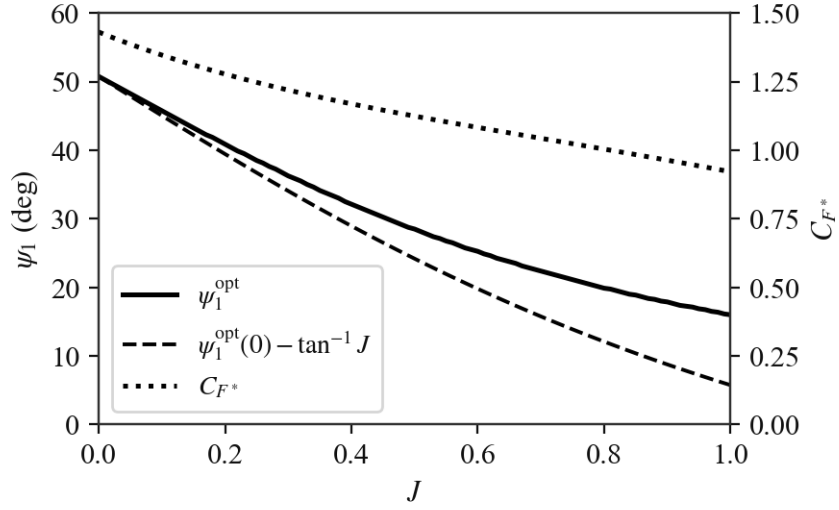


Figure 13: Pitch amplitude ψ_1^{opt} and force coefficient C_{F^*} of the continued hover optimum (the point tracked in Figure 12) as functions of advance ratio J . The dashed line is the approximation $\psi_1^{\text{opt}}(0) - \tan^{-1} J$, which holds the mid-stroke α constant as J increases.

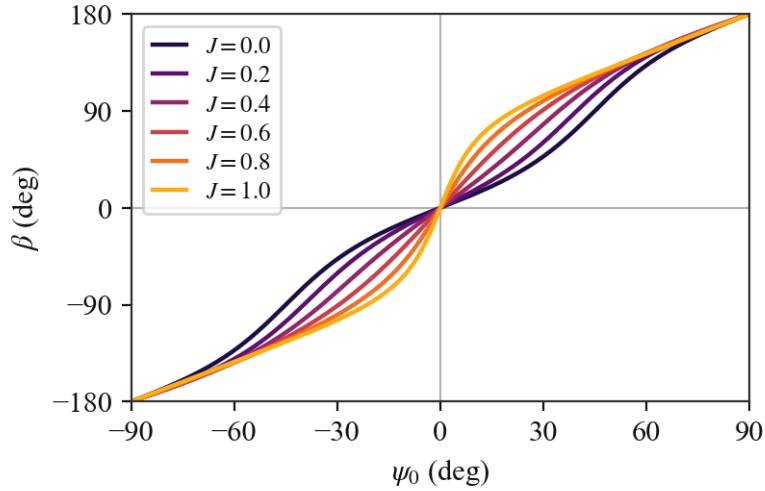


Figure 14: Effect of ψ_0 on the force direction angle β for increasing advance ratio J , at $\delta_0 = 90^\circ$ and with $\psi_1 = \psi_1^{\text{opt}}(J)$. Unlike in the hover case, β depends on ψ_1 when $J > 0$.

3.3 Control Strategy

From the above analysis, we propose a control strategy. It will be discussed qualitatively in this section, and formalized in the following section (subsection 3.4). We begin by going over *what to do* with each of the wing kinematic parameters.

- *Stroke plane angle γ* : when hovering, $\gamma = \gamma^{\text{hover}}$. When moving, γ should change to effectively obtain, or at least approximate, the axial inflow we have assumed in the analysis. There is the issue of the transition between these two phases, which may be abrupt if the commanded flight direction is not aligned with the stroke-plane normal at hover. We propose that γ be varied smoothly from its hover value to its moving flight value based on J .
- *Stroke amplitude s_0^** : primary control for the mean force magnitude, should be varied so that the force magnitude demand is met. In hovering or in general steady flight, this means having the mean force magnitude equal the weight.
- *Mean pitch ψ_0* : primary control for the mean force direction relative to the stroke-plane normal, should be varied so that the force direction demand is met. In hovering or in general steady flight, this means having the mean force vector point toward $+\hat{z}$.
- *Pitch amplitude ψ_1* : should be varied with J to maximize the available force, as shown in Figure 13.
- *Pitch phase δ_0* : fixed to 90° , not a control variable.
- *Wingbeat frequency ω^** : constant parameter, not a control variable.

In summary, the two variables that directly act to modulate and direct the force are s_0^* and ψ_0 . The two remaining variables, γ and ψ_1 , adapt to the velocity vector to respectively approximate an axial inflow and maximize the available force.

A comment should be made here about the $\hat{x}$ -wise asymmetry introduced by the nonzero γ^{hover} . At low speeds, when the stroke plane is still close to its hover orientation, this will result in different behaviors for motion toward $+\hat{x}$ and $-\hat{x}$. To restore symmetry, we propose to flip the sign of γ^{hover} when the commanded flight direction changes.

3.4 Control Law

3.4.1 Basic Formulation

We now formalize the strategy of subsection 3.3. Let $\vec{u}^*$ be the body velocity. The force demand is formed by a velocity tracking law. Given a reference velocity $\vec{u}_r^*$, the desired acceleration is proportional to the velocity error through a proportional gain K , and the force demand is obtained by adding a $+\hat{z}$ term which compensates for the weight:

$$\vec{a}_{\text{des}}^* = K (\vec{u}_r^* - \vec{u}^*) \quad (71)$$

$$\langle \vec{F}^* \rangle^{\text{des}} = \vec{a}_{\text{des}}^* + \hat{z} \quad (72)$$

A previous version of this report discusses augmenting Equation 71 with position feedback and acceleration feedforward terms to provide positional accuracy, but we do not pursue this here. The definition of the reference velocity $\vec{u}_r^*$ depends on the use case:

- *Hover*: $\vec{u}_r^* = \vec{0}$. The control law attempts to zero out the body velocity. See Figure 15.

- *Prescribed trajectory*: the trajectory is specified as a path $\vec{x}_r^*$ with a prescribed velocity $\vec{u}_r^*(\vec{x}_r^*)$ at every point, obtained as $\vec{u}_r^* = u_{\text{cmd}}^* \hat{t}$ where $\hat{t}$ is the local unit vector tangent to the path and u_{cmd}^* is the prescribed magnitude. $\vec{u}_r^*$ is fed to the control law at the same rate at which a point traveling along the path at exactly $\vec{u}_r^*$ would experience it. See Figure 16.
- *Prey pursuit*: the reference velocity is the prescribed pursuit speed u_{cmd}^* directed along the line of sight to the prey $\hat{r}$,

$$\vec{u}_r^* = u_{\text{cmd}}^* \hat{r} \quad \hat{r} = \frac{\vec{x}_{\text{prey}}^* - \vec{x}^*}{|\vec{x}_{\text{prey}}^* - \vec{x}^*|} \quad (73)$$

The control law attempts to steer the velocity vector towards the line-of-sight direction, similar to a proposed control law for tiger beetle prey pursuit [5] which attempts to minimize the angle between the velocity and line-of-sight vectors. See Figure 17.

The force demand is met by adjusting the control parameters, as discussed in subsection 3.4.3. The control law is exercised at the start of every wingbeat, so the control parameters remain constant through a wingbeat cycle and are only updated between cycles.

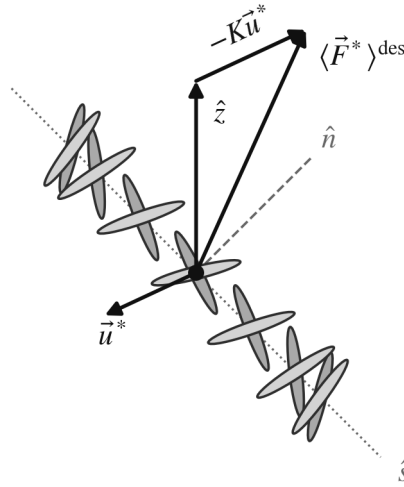


Figure 15: Control law geometry in the hover use case. The reference velocity is zero and therefore not drawn. The force demand of Equation 72 adds up the weight compensation $\hat{z}$ with the braking term $-K\vec{u}^*$ opposing the residual drift velocity $\vec{u}^*$, giving a resultant that supports the weight while arresting the drift. The stroke plane is fixed at $\gamma = \gamma^{\text{hover}}$ and $\psi_1 = \psi_1^{\text{opt}}(J = 0)$.

3.4.2 Gain Discussion

The gain K has a simple interpretation. Assuming the demanded mean force is actually delivered over one wingbeat cycle, the cycle-averaged equation of motion reduces to $\langle \vec{u}^* \rangle = K(\vec{u}_r^* - \vec{u}^*)$. This corresponds to first-order velocity tracking with time constant $1/K$. Let $\vec{e}_n$ be the velocity error sampled at the start of wingbeat n . Holding the demanded acceleration $K\vec{e}_n$ over the wingbeat period $T^* = 2\pi/\omega^*$ changes the velocity by $KT^*\vec{e}_n$, so, assuming the reference velocity remains constant over one beat,

$$\vec{e}_{n+1} = (1 - KT^*) \vec{e}_n \quad (74)$$

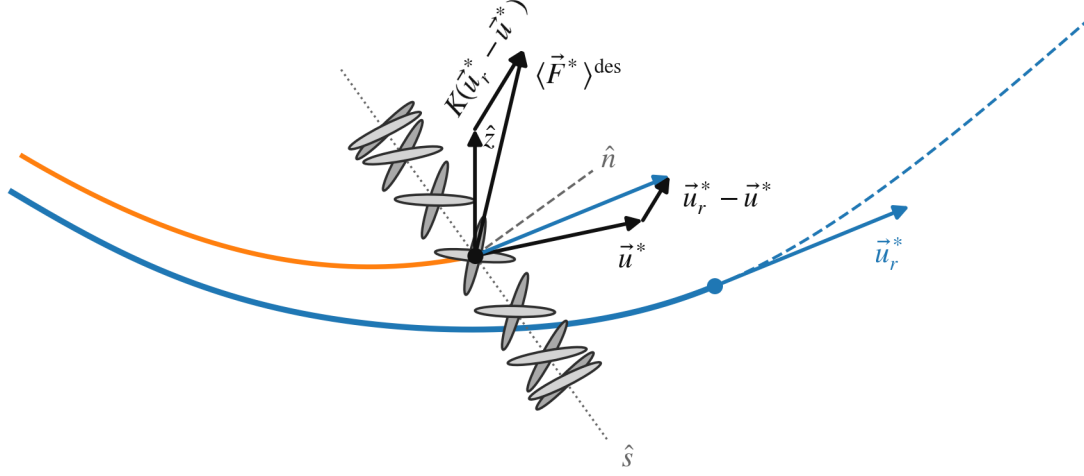


Figure 16: Control law geometry in the prescribed-trajectory use case. The reference point (blue dot) travels along the prescribed path (blue curve). The body is not constrained to the path since there is no position feedback, so its own path (in orange) does not line up exactly. The path velocity $\vec{u}_r^*$ shown in blue and the body velocity $\vec{u}^*$ give the error term $\vec{u}_r^* - \vec{u}^*$, and the force demand of Equation 72 adds up the weight compensation $\hat{z}$ with the amplified error $K(\vec{u}_r^* - \vec{u}^*)$.

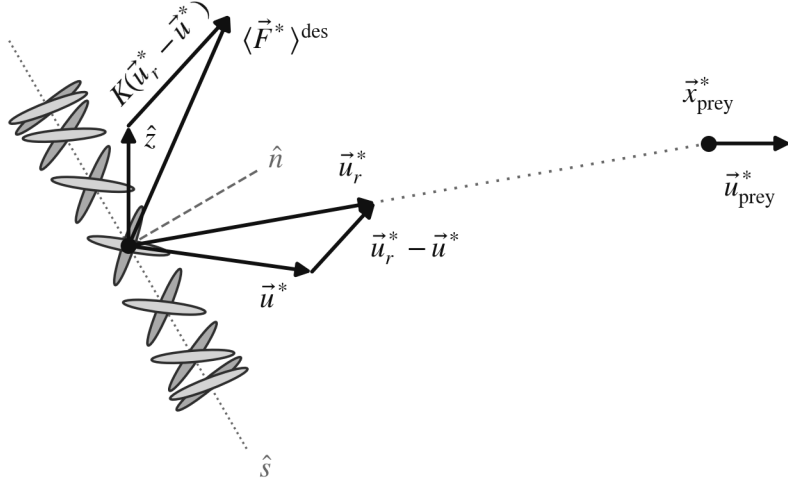


Figure 17: Control law geometry in the prey pursuit use case. The prey is detected at the line-of-sight bearing $\hat{r}$ (sparse dots), and the reference velocity defined as $\vec{u}_r^* = u_{\text{cmd}}^* \hat{r}$ (Equation 73) points along it. The control law attempts to correct the velocity error $\vec{u}_r^* - \vec{u}^*$ to maintain the pursuit course. The force demand of Equation 72 adds up the weight compensation $\hat{z}$ with the amplified error $K(\vec{u}_r^* - \vec{u}^*)$. The prey position $\vec{x}_{\text{prey}}^*$ enters the control law only to obtain $\hat{r}$, with all range information discarded. The prey velocity $\vec{u}_{\text{prey}}^*$ never enters the control law.

The factor $(1 - KT^*)$ represents the decay (or amplification) of the velocity error over one cycle. The convergence behavior is standard (see e.g. [6]): monotone convergence for $KT^* < 1$, single-beat (so-called *deadbeat*) convergence at $KT^* = 1$, where the beat delivers exactly the velocity increment that cancels the sampled error, and sign-alternating convergence up to the stability bound $KT^* = 2$, with divergence beyond. Here, Equation 74 is an approximation: even if the force demand is perfectly met for the conditions at the start of the wingbeat, body velocity fluctuations during the wingbeat will cause it to deviate to some extent. The reference velocity itself may also have changed by the end of the wingbeat. Nevertheless, given the relatively short timescale of the wingbeat cycle compared to the body motion (an assumption that the mean aerodynamic force calculation rests upon), we consider it a useful approximation. Since it provides the fastest theoretical response, we select $K = 1/T^* = \omega^*/2\pi$ in the simulations presented below.

3.4.3 Achieving the Demanded Force

With the demanded force formulated, the four variables γ , ψ_1 , ψ_0 , and s_0^* must be set to effectively achieve it. First, the ‘passive’ variables γ and ψ_1 are adapted to the new body velocity. For this purpose, the advance ratio is evaluated as

$$J = \frac{|\vec{u}^*|}{(s_0^* \omega^*)^{\text{ref}}} \quad (75)$$

where $(s_0^* \omega^*)^{\text{ref}}$ is the wingbeat velocity of the reference configuration at the hover condition, such that J is independent of the control variable s_0^* , to prevent an undesirable feedback loop. At high J , γ is made to approach the velocity-aligned value γ_u , defined such that $\hat{n}$ is aligned with $\vec{u}^*$, as $\gamma_u = \text{atan2}(-u_x^*, u_z^*)$. This gives $\mp 90^\circ$ in level flight toward $\pm \hat{x}$, 0° in climb, and $\pm 180^\circ$ in descent. γ blends from γ^{hover} to γ_u as

$$\gamma = [1 - f(J)] \sigma_x \gamma^{\text{hover}} + f(J) \gamma_u \quad f(J) = 1 - e^{-(J/J_c)^2} \quad (76)$$

with a somewhat arbitrary $J_c = 0.35$, placed between the two velocity regimes. Tuning of J_c has not been pursued yet. At small body velocities ($J \lesssim 0.1$) the stroke plane effectively remains in its hover inclination. At cruising speeds ($J \gtrsim 2J_c$) the blend is essentially complete ($f > 0.98$). The sign $\sigma_x \in \{-1, +1\}$ sets the tilt direction of the hover stroke plane (subsection 3.3): it is the sign of the horizontal component of the commanded velocity, $\sigma_x = \text{sgn}(\vec{u}_r^* \cdot \hat{x})$. σ_x is latched (retaining its previous value whenever $\vec{u}_r^* \cdot \hat{x} = 0$) and initialized to +1. Figure 18 illustrates the change in γ for accelerating level flight.

A notable issue with this γ mechanism is descent. During descent, γ_u aims the stroke-plane normal $\hat{n}$ downward. If the prescribed velocity is steady, rather than accelerating, the weight must be supported, which then requires a mean force along $-\hat{n}$. As established in subsection 3.2 (Figure 12), such a reversed force is reached at mean pitch near $\pm 90^\circ$, where drag from the normal inflow dominates. This has proven to be a delicate regime, and so the current formulation is limited in descent to low velocities (or sustained accelerations greater than that due to gravity). An alternate control scheme making passive use of drag to control descent velocity could perhaps be devised.

The other passive variable, the pitch amplitude ψ_1 , is slaved to the optimal value

$$\psi_1 = \psi_1^{\text{opt}}(J) \quad (77)$$

where $\psi_1^{\text{opt}}(J)$ maximizes the stroke-normal coefficient $C_{F_n^*}$ of Equation 63 at neutral mean pitch $\psi_0 = 0^\circ$.

With (γ, ψ_1) now set, the remaining two variables (s_0^*, ψ_0) are set to satisfy the force demand by solving

$$\langle \vec{F}^* \rangle (s_0^*, \psi_0; \gamma, \psi_1, \vec{u}^*) = \langle \vec{F}^* \rangle^{\text{des}} \quad (78)$$

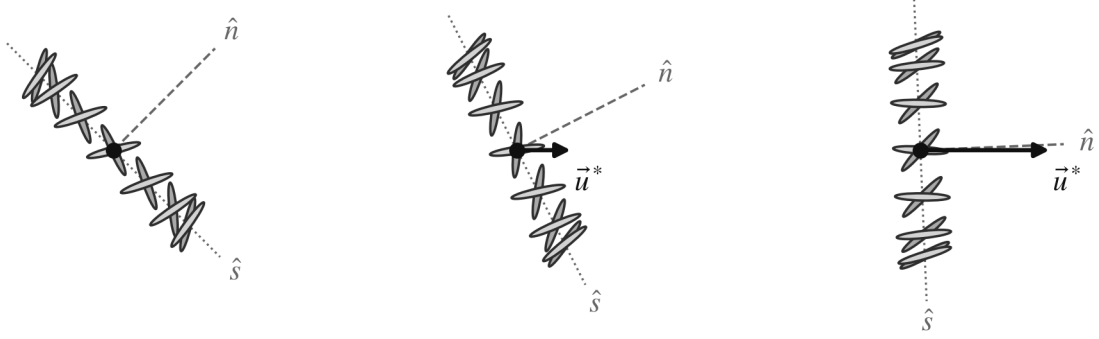


Figure 18: Stroke plane tilting (Equation 76) in level flight toward $+\hat{x}$ ($\sigma_x = +1$), at increasing advance ratio. At rest (left), the stroke plane holds the hover stroke plane angle $\sigma_x \gamma^{\text{hover}}$. As the forward speed builds (middle, right), the stroke plane tilts towards the velocity-aligned value γ_u , at which the stroke-plane normal $\hat{n}$ points along the flight direction and the inflow is axial.

The two unknowns match the two numbers that the planar force demand supplies once resolved in the stroke frame: direction and magnitude. The direction is the signed angle β^{des} from the stroke-plane normal $\hat{n}$ to $\langle \vec{F}^* \rangle^{\text{des}}$. The magnitude is $F^{\text{des}} = |\langle \vec{F}^* \rangle^{\text{des}}|$. Both are known once γ is set.

The mean force at the operating point, $\langle \vec{F}^* \rangle = q^* (C_{F_n} \hat{n} + C_{F_s} \hat{s})$ with the coefficients of Equation 63 and Equation 64, splits these two roles cleanly between the two unknowns. Because J is evaluated at the fixed reference wingbeat velocity, the coefficients are independent of s_0^* , which enters only through $q^* \propto (s_0^* \omega^*)^2$ (Equation 38), so s_0^* scales the mean force without turning it. The direction β is in turn governed by ψ_0 alone (Equation 68), while the magnitude depends on both unknowns. Equation 78 can therefore be split into

$$\beta(\psi_0; \psi_1, J) = \beta^{\text{des}} \quad q^* C_{F^*}(\psi_0; \psi_1, J) = F^{\text{des}} \quad (79)$$

solved in that order. The direction demand fixes ψ_0 by inverting Equation 68, and the magnitude demand then yields the amplitude explicitly through q^* (Equation 38)

$$s_0^* = \frac{1}{\omega^*} \sqrt{\frac{4F^{\text{des}}}{C_{F^*} \frac{A^*}{m^*}}} \quad (80)$$

with C_{F^*} evaluated at the new ψ_0 and ψ_1 using Equation 63 and Equation 64. This is the generalization of the hover condition of subsection 3.2, recovered when the body is at rest and the demand reduces to the weight: $\beta^{\text{des}} = -\gamma^{\text{hover}}$ returns ψ_0^{hover} , and $F^{\text{des}} = 1$ returns $(s_0^*)^{\text{hover}}$.

In practice, the expressions we derived assuming normal inflow are only adequate about either the hover condition (low velocity) or when the stroke-plane normal $\hat{n}$ is actually closely aligned with the velocity, which is generally the case for pursuit. As a result, both the hover and pursuit force demands can be adequately solved in the manner described above, as shown in section 4. For more complex trajectories with tight maneuvers, which are likely to result in large transient deviations of the inflow angles from the ideal normal case, the above method tends to fail unless u_{cmd}^* is very small. In such cases, a numerical solve using a Newton iteration on the residual of Equation 78, with the force obtained using the general form (Equation 5), without any assumption on the inflow angle, has proven necessary to achieve adequate trajectory tracking. Convergence occurs in a handful of steps and each solve is warm-started from the previous wingbeat's (s_0^*, ψ_0) , since the demand moves little between solves. This numerical solve method is used for the prescribed trajectory test cases, while the analytical method is used for hover and prey pursuit. Using the simplified constant-pitch

kinematics for the analytical derivation could allow us to obtain a general expression for any inflow angle, removing the need for the numerical solve. Furthermore, by linearizing this result, the control law could be simplified and expressed directly with the control variables.

4 Results

4.1 Reference Configuration

To ground our results, we introduce a reference configuration based on morphological and wingbeat frequency data for *Anisoptera* from [7] and [8], respectively. Nondimensionalized, the data range is $[0.2, 1.0]$ for A^*/m^* , and $[8, 20]$ for ω^* . For the reference configuration, we select the intermediate values

$$\frac{A^*}{m^*} = 0.6 \quad \omega^* = 14 \quad (81)$$

In addition, we choose the hovering stroke plane angle to be

$$\gamma^{\text{hover}} = -45^\circ \quad (82)$$

defined with a negative sign (clockwise) so that it corresponds to a horizontal dragonfly pointing towards $+\hat{x}$. When settling from lateral motion towards $-\hat{x}$, the hovering stroke plane angle would be $-\gamma^{\text{hover}} = +45^\circ$, as explained in [subsection 3.3](#). Since dragonflies have two wing pairs, we set $N = 2$, with the two pairs flapping out of phase (their phase parameters ϕ differing by 180°) so as to reduce body oscillations.

4.2 Hover

The hover regime of the control law is exercised by initializing the body at the origin with a velocity perturbation $\vec{u}^*(0) = (0.2, 0.2)$, and $\vec{u}_r^* = 0$, with $KT^* = 1$. The response is shown in [Figure 19](#).

4.3 Prescribed Trajectory

Results for two reference trajectories are shown in [Figure 20](#) and [Figure 21](#). The first is designed to exhibit a variety of features including tighter and wider turns, loops, and straight lines in all orientations, while the second is a freehand trajectory. Both trajectories are traversed with a trapezoidal speed schedule $\vec{u}_r^*(t)$ made of linear ramps from and to rest over a distance of 0.3. u_{cmd}^* is kept moderate (0.5 and 0.35 for the first and the second trajectory, respectively) because of the vertical descent segments, which fail at high descent rates ([subsubsection 3.4.3](#)). At the relatively low speeds shown, trajectory tracking is fairly adequate and the achieved trajectories closely match the originals. These trajectory tracking cases rely on the numerical solve method discussed at the end of [subsubsection 3.4.3](#).

4.4 Prey Pursuit

Prey pursuit closes the guidance loop through the control law with the reference velocity of [Equation 73](#), giving the desired pursuit speed along the line of sight to the prey. Here the prey starts above and to the right, at $(4, 4)$, and moves at the constant velocity $(1, 0)$; the body starts at rest at the origin, with $u_{\text{cmd}}^* = 1.5$ and $KT^* = 1$. Interception is declared at a range of 0.5 body lengths. The resulting pursuit is shown in [Figure 22](#), and exhibits the classical pure-pursuit form, albeit with some overshoot.

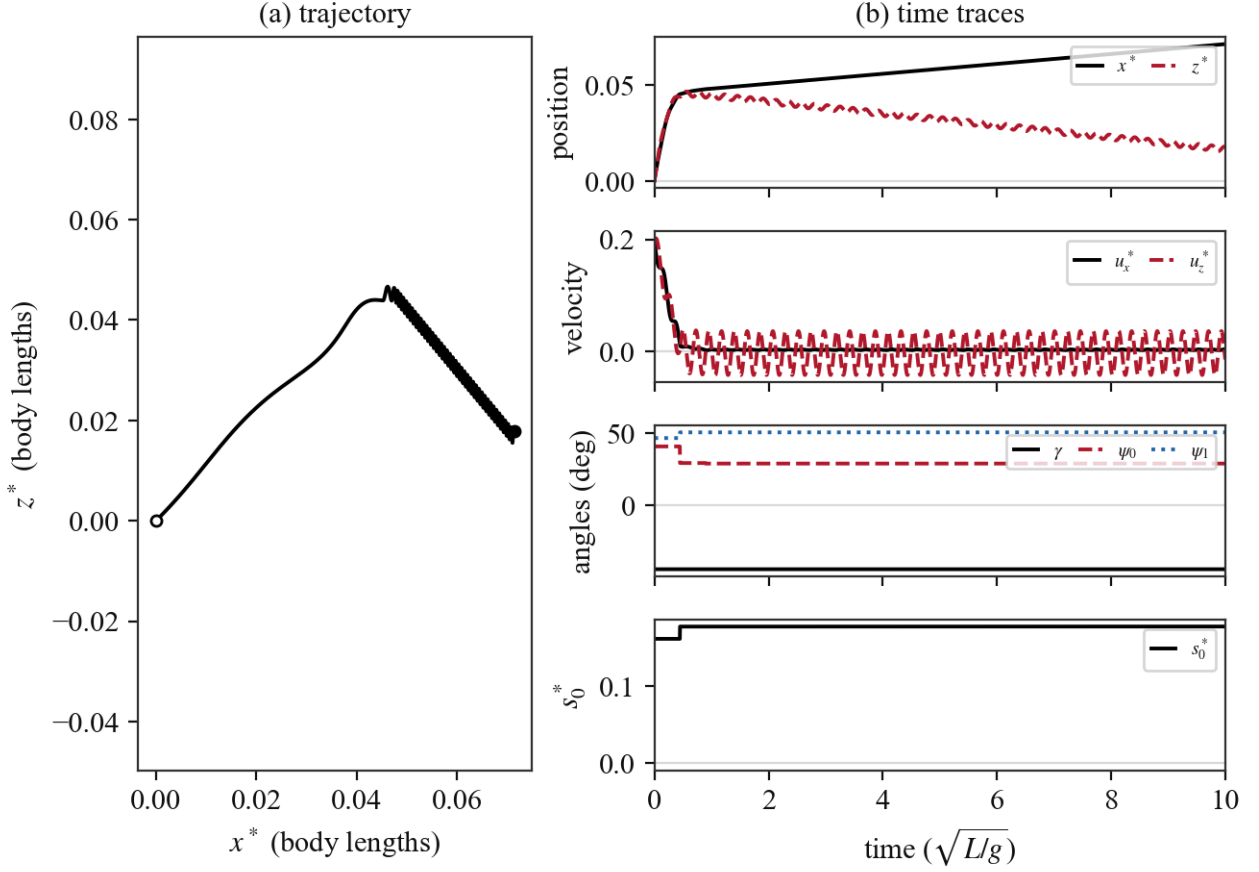


Figure 19: Hover control with initial velocity perturbation at $KT^* = 1$. The final hover values of the control parameters are reached effectively after the first wingbeat, and the velocity perturbation is eliminated within a few wingbeats. The small position offset accumulated meanwhile is never corrected since there is no position feedback, and some slow residual drift remains, but the body settles into a clean hover about its final position.

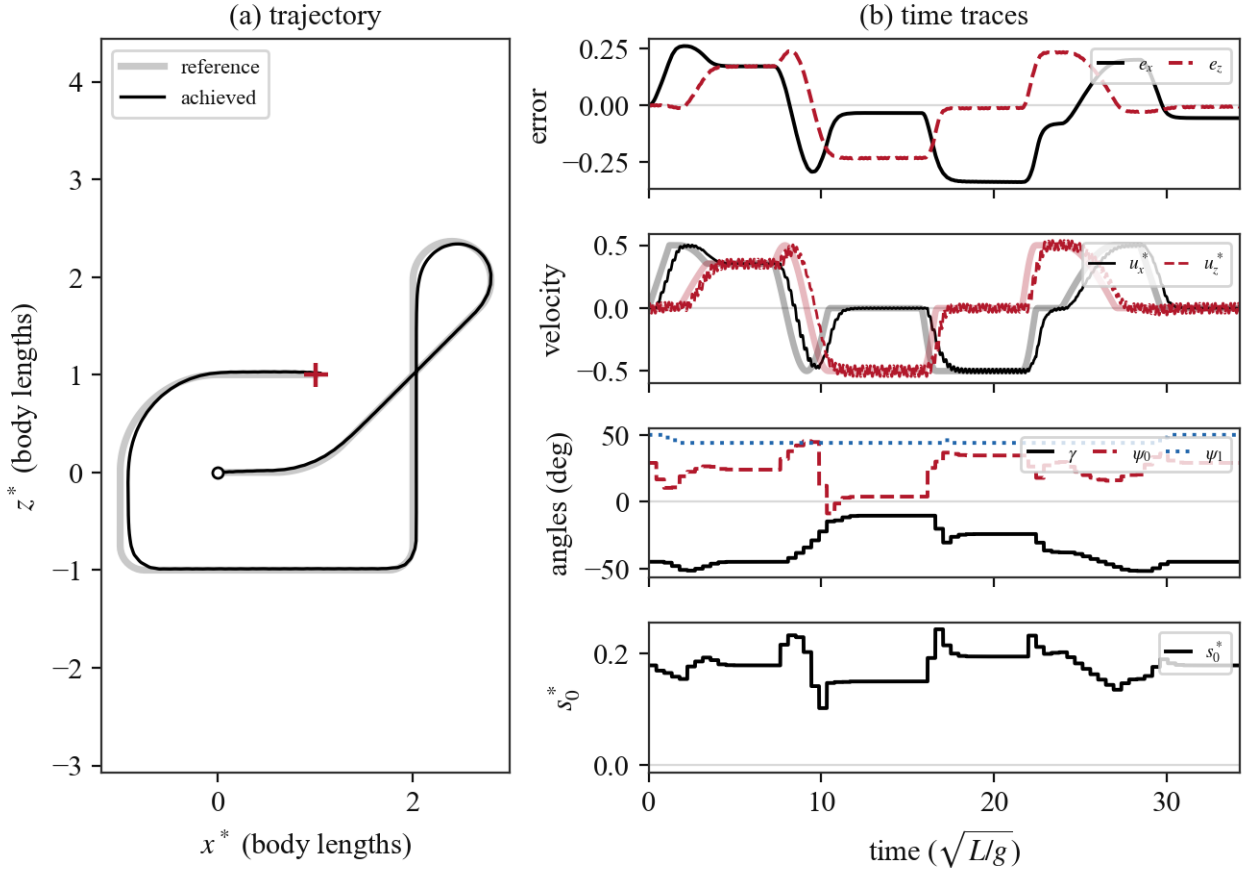


Figure 20: Prescribed trajectory with $u_{\text{cmd}}^* = 0.5$ and $KT^* = 1$. The body follows the commanded velocity with a first-order lag, trailing the reference point by $|\vec{u}_r^*|/K \approx 0.22$ body lengths through the cruise and cutting slightly inside the sharp corners, with the tracking error peaking at 0.34. Position drift is not corrected, but the achieved path stays close to the course throughout, including the loop and the descent, and the body settles next to the course endpoint (final offset 0.06). This same trajectory fails on descent when $u_{\text{cmd}}^* = 1$.

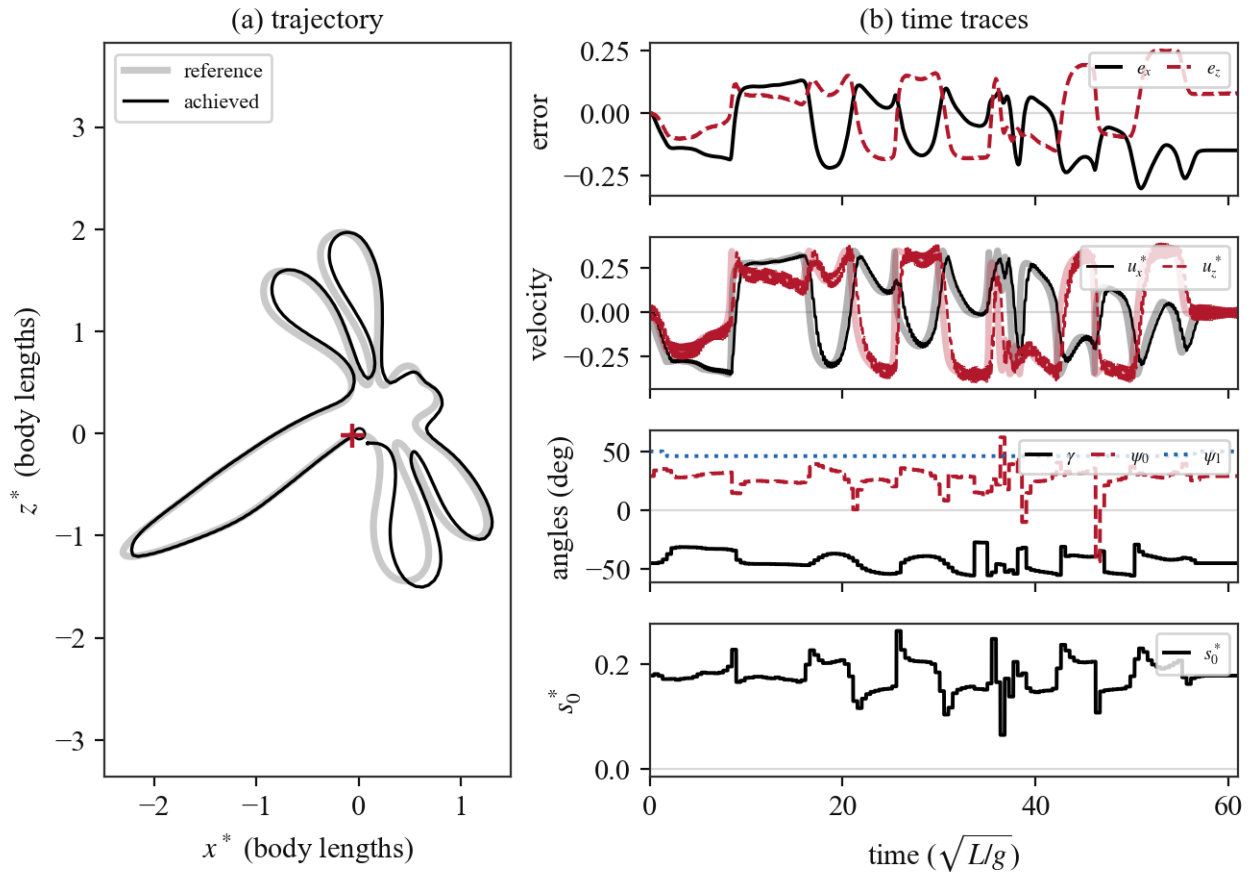


Figure 21: Freehand trajectory with $KT^* = 1$ and $u_{\text{cmd}}^* = 0.35$.

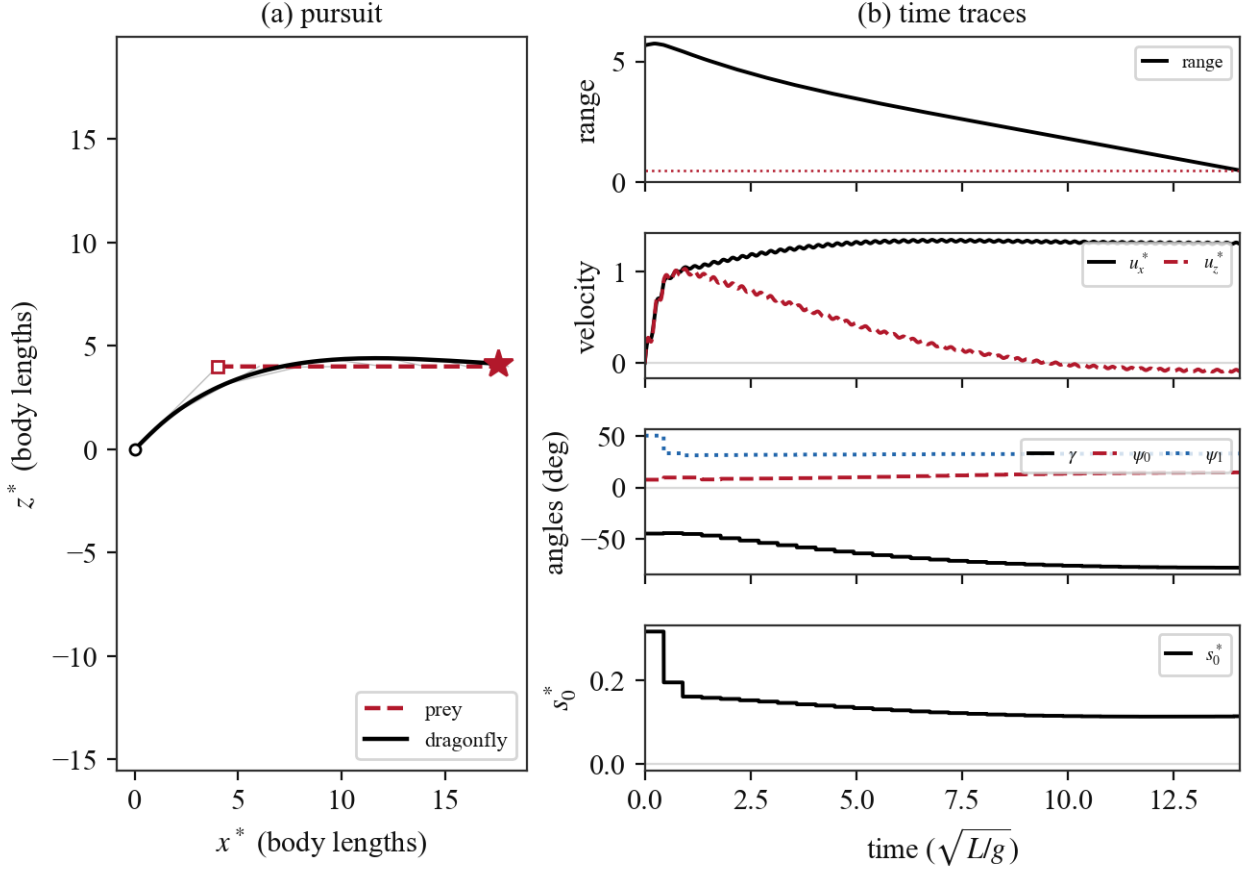


Figure 22: Prey pursuit with $u_{\text{cmd}}^* = 1.5$, $\vec{u}_{\text{prey}}^* = 1\hat{x}$, and $KT^* = 1$. The line-of-sight command produces a classical pure-pursuit trajectory, with successful capture. The body accelerates to the commanded pursuit speed and γ leans from γ^{hover} toward alignment with the flight direction, as ψ_1 follows its slaved schedule and (s_0^*, ψ_0) adjust to the force demand.

5 Conclusion

The proposed control scheme is capable of achieving hover and prey pursuit, and of following arbitrary trajectories to a reasonable degree of accuracy at moderate velocities. Some caveats are worth mentioning:

- *Not strictly proportional control.* While all control tasks can be achieved with a single proportional gain K in the force demand formulation, the control variables vary nonlinearly with the force demand, since we are solving the inverse problem of which variable values satisfy the force demand. We may be able to resolve this by 1) obtaining closed-form expressions for the mean aerodynamic force using the simplified, constant-pitch kinematics and 2) linearizing those expressions.
- *Issue with descent.* When descending (towards $-\hat{z}$) at a fast rate, the stroke-plane normal will point downwards. However, to achieve a steady descent velocity, the force vector must point up, opposed to gravity. The force vector, then, must point toward $-\hat{n}$. This can be achieved, as we saw in [subsection 3.2](#), with large enough $|\psi_0|$, but at high J it results in large force magnitudes which lead to instability in the unsteady dynamics, and prevent clean descent trajectories. This could potentially be addressed with an alternate control mode using gliding instead of flapping to achieve steady descent. Note that descent is not as problematic if the commanded trajectory has a sustained downward acceleration larger than that due to gravity, since the requirement that the force oppose gravity does not hold.

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